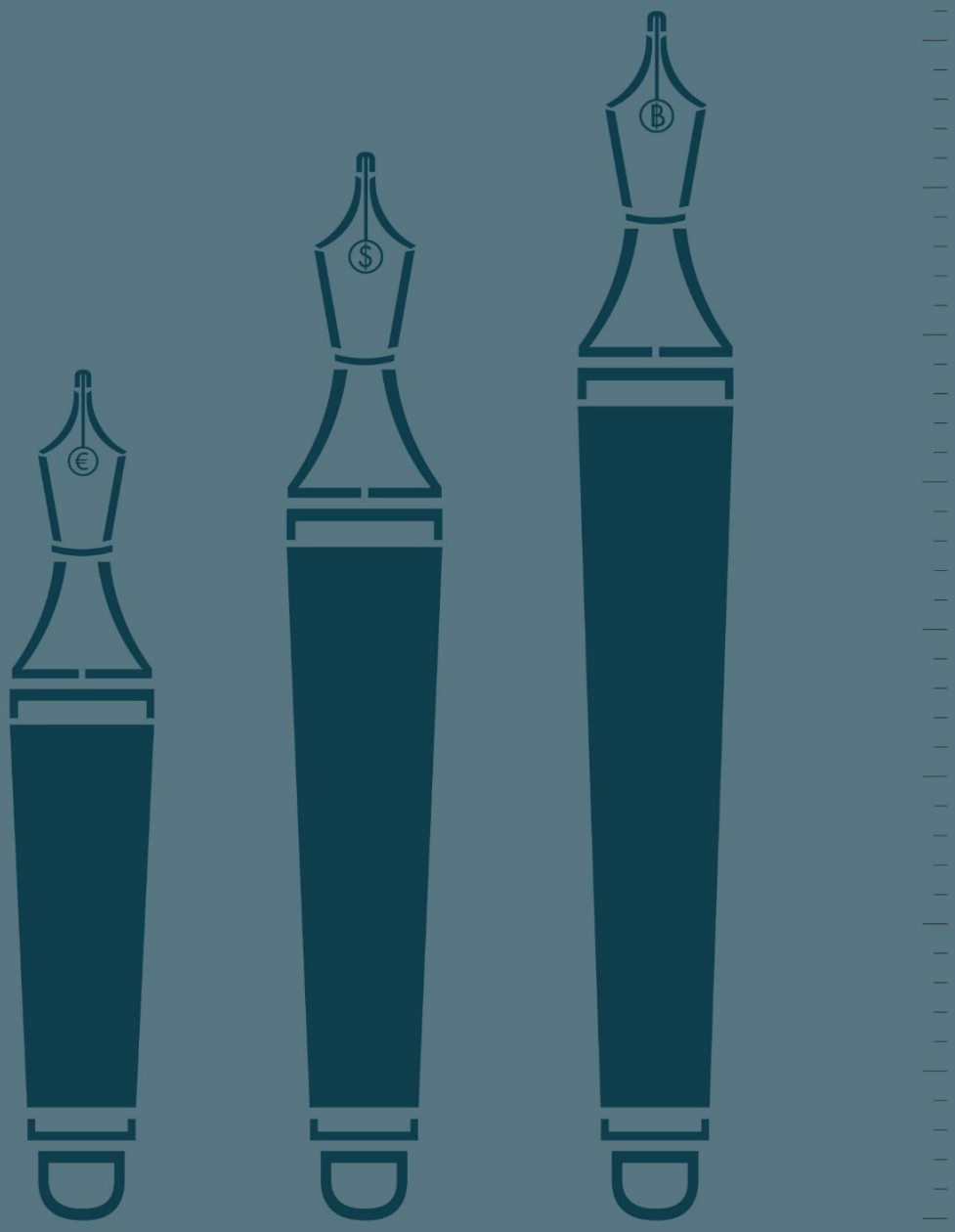
BANK OF THAILAND

MONETARY POLICY REPORT SEPTEMBER 2016





BANK OF THAILAND

Monetary Policy Report

September 2016

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
September 2016

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Jamlong Atikul	Member
Mr. Porametee Vimolsiri	Member
Mr. Apichai Boontherawara	Member
Mr. Sethaput Suthiwart-Narueput	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for 2016 and for the medium term. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

Monetary Policy Report

September 2016

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Growth and Inflation Prospects and Monetary Policy

1. Growth and Inflation Prospects and Monetary Policy

The Thai economy is projected to expand slightly faster than assessed in the previous *Monetary Policy Report* thanks to the higher-than-expected growth in private consumption in the second quarter of 2016. This positive development helps compensate for the contraction in merchandise exports owing to a projected slowdown in trading partners' economies post-Brexit and Thailand's own structural problems in the export sector. For 2017, the Thai economy is expected to grow at about the same rate as this year, driven mainly by private consumption and public spending and also by expansion in tourism.

Core inflation in 2016 and 2017 is expected to remain unchanged from the previous estimate. Increased demand pressures from the economic recovery are offset by a decline in costs because of lower oil prices resulting from weaker global demand. Headline inflation is revised down in line with a downward revision in the crude oil price forecast. While the Committee expects headline inflation to return to the lower bound of the target band within this year, the timing would largely depend on movements in global crude oil prices.

The Committee maintained the policy interest rate in the August and September meetings. The overall economic outlook remains largely unchanged from the meetings in the preceding quarter. The economy is expected to recover at a gradual pace and inflation is projected to return to the target band, with monetary conditions continuing to be accommodative. While a potential increase in risk accumulation under the prolonged low interest rate environment continues to warrant close monitoring, financial stability remains sound. Nevertheless, after Brexit the Thai economy is faced with greater risks on the external front, which will affect the outlook of Thailand's trading partners and overall confidence. While there is a high degree of uncertainty associated with these risks, their consequences can be severe should they materialize. The Committee thus emphasizes the need to preserve policy space, continuing to monitor and assess risks to the economic outlook in order to formulate appropriate monetary policy.

1.1 Growth and inflation prospects

The Committee revises the GDP growth forecast for 2016 from 3.1 to 3.2 percent **(Table 1.1)** mostly on account of **higher-than-expected growth in the first half of the year** that was particularly contributed by private consumption. The strong consumption growth is due to the easing of the drought, together with temporary boosts from government stimulus measures to stimulate private consumption as well as accelerated purchases of durable goods following the launch of new car models and promotional campaigns. **Although growth is likely to moderate somewhat in the second half of the year from the first half with short-term government stimulus measures expiring, the economy is expected to continue to expand albeit at a gradual pace**. Growth will be supported by the gradual improvement in farm income, as the impact of the drought subsides, and also driven by disbursement of public spending that has been well-maintained. These positive factors will help underpin private spending, thereby providing support for the overall economy given that merchandise exports have yet to recover and exports of services experience a setback in the short term from the government's attempt to curb zero-dollar tours and the bombing incidents in the seven provinces in Southern Thailand in August.

The Thai economy is projected to grow in 2017 at about the same rate as in this year. Domestic demand will remain a key driver of growth on the back of continuity of government policies after the draft constitution was approved in the referendum. In particular, new investment in transportation infrastructure such as the dual-track railway and Bangkok Mass Transit projects are

Table 1.1 Forecast summary

Percent	2015*	2016	2017
GDP growth	2.8	3.2	3.2
		(3.1)	(3.2)
Headline inflation	-0.9	0.3	2.0
		(0.6)	(2.2)
Core inflation	1.1	0.8	1.0
		(0.8)	(1.0)

Note : *Outturn
() June 2016 MPR
Source : Office of National Economic and Social Development Board, Ministry of Commerce, calculations by Bank of Thailand

expected to start in the second half of the year. Tourism is expected to return to normal after the adverse impacts of explosions in the southern provinces and the crackdown on zero-dollar tour operators diminish. Nonetheless, growth momentum of merchandise exports is likely to remain limited and will continue to weigh on the economic recovery given weaker-than-expected growth in trading partners' economies and structural problems in Thailand's export sector. **The Committee therefore decided to maintain the growth forecast for 2017 at 3.2 percent (Table 1.1).**

Overall macroeconomic improvements in 2016 and a continued expansion in 2017 will induce higher demand pressures as reflected by the narrowing of the output gap over the forecast period **(Chart 1.1)**. Meanwhile, cost-push inflationary pressures decline, as lower crude oil prices and domestic energy prices are transmitted to other goods and services. The Committee therefore maintains the core inflation forecast for 2016 and 2017 at 0.8 and 1.0 percent, respectively. However, the Committee revises down the headline inflation forecast or 2016 and 2017 to 0.3 and 2.0 percent, respectively, from the previous assessment of 0.6 and 2.2 percent. Nonetheless, **headline inflation is still projected to return to the lower bound of the target band within this year**, the timing of which, however, depends mainly on global oil price movements.

Key economic developments that the Committee takes into account in preparing the growth and inflation forecasts are as follows.

(1) Trading partners' economies are expected to grow at a slower rate than previously assessed (Table 1.2), continuing to

Chart 1.1 Output Gap

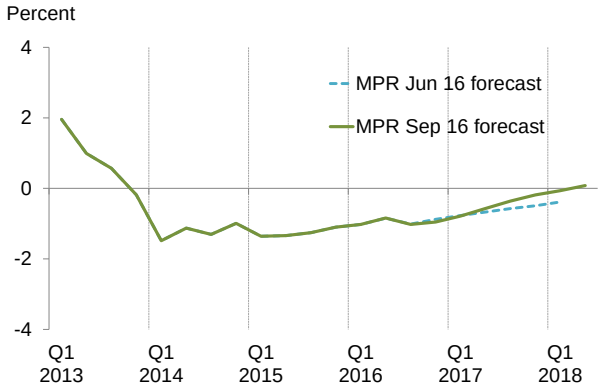


Table 1.2 Growth assumptions for Thailand's trading partners

Percent (%YoY)	Weight (%)	2015	2016		2017	
			Jun 16	Sep 16	Jun 16	Sep 16
United States	14.9	2.4	1.9	1.5	2.3	2.2
Euro area	10.0	1.6	1.7	1.6	1.6	1.2
United Kingdom	2.5	2.3	1.8	1.6	2.2	0.6
Japan	13.6	0.6	0.6	0.5	1.0	0.9
China	15.7	6.9	6.5	6.5	6.4	6.3
Asia (ex Japan and China)*	37.4	3.5	3.2	3.2	3.7	3.5
Total**	100	3.2	3.0	2.9	3.3	3.1

Note: * Weighted by each trading partner's share of Thailand's total exports in 2014, namely Singapore (6.5%), Hong Kong (7.9%), Malaysia (8%), Taiwan (2.5%), Indonesia (5.9%), Korea (2.8%), and the Philippines (3.7%)
** Weighted by each trading partner's share of Thailand's total exports as of 2014 (13 countries)

weigh on Thailand's merchandise exports over the period ahead.

The Committee revises down projections of trading partners' growth for 2016 and 2017 on account of (1) lower-than-expected second-quarter GDP outturns, especially for the US, (2) the UK referendum to leave the EU (Brexit) that will likely restrain growth in the euro area amid falling consumer and business confidence due to increased political uncertainties, and (3) impacts from heightened volatility in the global financial markets which likely lead to the appreciation of the US dollar and the Japanese yen. Such appreciation will weigh on the economic recovery of the US and Japan, which in turn will affect the Chinese and Asian economies.

The lower-than-expected trading partners' growth is an important factor that restrains Thai exports in the period ahead, further exacerbating the contraction of most merchandise exports. However, the value of gold exports, which has risen on the back of the recent increases in gold prices post-Brexit, will help compensate for the contraction of other merchandise exports. **The Committee therefore maintains the forecast of export value at -2.5 percent in 2016. For 2017, the Committee revises down** the export value from zero growth **to a contraction of 0.5 percent in anticipation of increasingly greater impacts from Brexit going forward (Table 1.3 in the Appendix to Chapter 1).** The subdued outlook for merchandise exports suggests low growth for production and employment in export-oriented manufacturing industries, thereby affecting household spending and business investment decisions.

Moreover, major advanced economies are likely to maintain very accommodative policy for a

longer period of time ("low for longer") given the slow global economic recovery. The Federal Reserve is expected to continue monetary policy easing for longer than previously assessed while awaiting to further evaluate global financial conditions and the impact of Brexit before being expected to raise the federal funds rate in December 2016, as opposed to September 2016 in the previous assessment. Meanwhile, the Bank of England eased monetary policy further at the August meeting by cutting the policy rate, purchasing government and corporate bonds, and introducing the Term Funding Scheme (TFS). The Bank of Japan also continues with monetary policy easing through the negative interest rate policy and Quantitative and Qualitative Easing (QQE). The uncertainty in monetary policy directions of major advanced economies would further contribute to capital flow and exchange rate volatility in the period ahead.

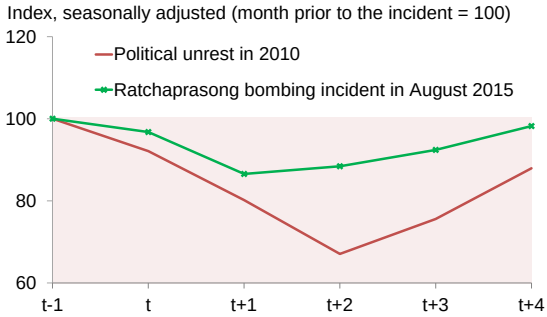
The Committee assesses risks to growth of Thailand's trading partners to be tilted to the downside due to greater uncertainty post-Brexit. In particular, the trade and investment agreement negotiations between the UK and the EU will have significant implications for global trade and long-term global economic recovery. In addition, the following risks to trading partners' economic recovery warrant close monitoring: (1) uncertainty in monetary policy directions of major advanced economies; (2) political uncertainty in the US with the upcoming presidential election in November; (3) possible increase in political uncertainty in Europe after elections in several countries especially France, Italy, and the Netherlands, and (4) risks in the Chinese financial sector from high corporate debt levels. These risks can heighten global financial volatility and may affect the real economy more than expected.

(2) **Growth momentum from tourism may be more modest than expected, especially in 2016**, due to (1) the actual number of foreign tourists in the second quarter grew less than expected; (2) trading partners' growth was also lower than previously assessed, although the impact was likely limited to European tourists and less so for Chinese tourists who constitute a major share in the tourist market; (3) short-term setback in the number of tourists after the bombing incidents in August, although a rapid rebound is expected as with other incidents in the past **(Chart 1.2),** and (4) the government's attempt to curb zero-dollar tours that aims to regulate Chinese tour operators, which can lead to some decline in the number of Chinese tourists in the fourth quarter in 2016 and the first quarter in 2017. Consequently, the Committee projects the number of foreign tourists in 2016 and 2017 to be 33.6 and 36.3 millions, slightly down from the previous estimate at 34.0 and 36.7 millions, respectively.

Furthermore, the tourism sector faces additional short-term risks stemming from (1) the recent bombing incidents that may deteriorate tourist confidence more than expected and (2) regulation to set a minimum price for tour packages which is expected to come into effect in 2017 that may trim down the projected number of Chinese tourists who constitute the major tourist group for Thailand, constituting a significant risk going forward.

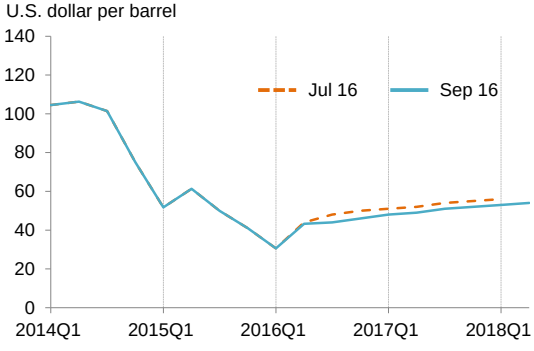
(3) **The increase in global crude oil prices is likely to be slower than expected throughout the assessment period (Chart 1.3), leading to a downward revision of Thailand's headline inflation**. The Committee revises down the assumption on Dubai oil price over the forecast period, from the average of 41.0 and 50.0 US

Chart 1.2 Index of foreign tourists during shocks in the past



Source: Department of Tourism and calculations by the Bank of Thailand

Chart 1.3 Assumptions on Dubai oil price



dollars per barrel in 2016 and 2017 respectively from the previous assessment of 43.1 and 53.0 US dollars per barrel, respectively. This is on account of weak global demand from the slower-than-expected global economic recovery. Concurrently, supply conditions remain unchanged from the previous assessment as OPEC production is expected to slow down slightly due to production outages in many countries, while production in Non-OPEC countries remains low despite the fact that shale oil producers in the US are likely to raise production levels as prices gradually increase.

Lower oil prices compared with previous projection have allow domestic costs of goods and services, especially in transportation, to stabilize at a low level, resulting in lower-than-expected headline inflation. Moreover, **low oil prices also suppress prices of oil-related exports and imports.** Given that oil imports constitute a significant share of Thailand's total imports, together with the fact that import of capital goods are lower compared with the previous *Monetary Policy Report*, the value of imports declines more than that of exports. As a result, **the current account in 2016 and 2017 is expected to record a larger surplus of 40.4 and 31.8 billion U.S. dollars**, up from the previous estimates of 37.8 and 32.3 billion U.S. dollars, respectively.

Looking ahead, **global oil prices are posing larger downside risks to the baseline forecast.** Factors that may lead to an underestimation of oil prices include (1) lower-than-expected global oil demand due to a slow global economic recovery, (2) crude oil supply from shale oil producers that may rebound more quickly than expected once prices are higher, and (3) crude oil supply from major oil exporting countries that may

be larger than expected as they increase production to maintain market shares. On the other hand, upside risk affecting the baseline forecast may stem from (1) unrests in the Middle East that can spread to major oil production areas and (2) negotiations between OPEC and non-OPEC producers in September. If an agreement on production freeze is reached, prices will likely rise in the future.

(4) Public sector disbursement is higher than previously assessed following the referendum result to accept the draft constitution. Government expenditure in 2016 is slightly lower than predicted as the outturn in the second quarter shows a lower-than-expected investment by state-owned enterprises (SOEs). The total figure for 2016 can be partly compensated for by the government's measure to expedite budget disbursement in the last quarter of this year. In 2017, expedited carry-over expenditure in accordance with the Transfer of Expenses Budget Act, B.E. 2559 will continue supporting robust public expenditure. Moreover, the approval of the draft constitution will allow the government to push through investment projects with a clearer timeline, especially those in transportation infrastructure whose projects are scheduled to launch in the beginning of 2017 such as the dual-track rail and Bangkok Mass Transit projects. Increased government expenditure, an upward trend in public investment, and the materialization of the public-private partnership (PPP) in the Pink Route and Yellow Route rail construction projects will encourage confidence and support private investment in 2017.

(5) Recent economic developments above are key factors in supporting **a gradual recovery of private consumption**. In the second half of

2016, private consumption growth may slow down somewhat after having accelerated in the second quarter. In 2017, private consumption is expected to recover slowly supported by (1) robust income growth in the service sector, (2) farm income that is improving but remain at a low level, and (3) lower costs of living due to lower oil prices. Meanwhile, the recovery of income in export-oriented manufacturing sectors remains restrained by weaker global growth and structural limitations facing the Thai export sector.

(6) Private investment projection is lower than expected due to (1) subdued merchandise exports, and (2) the economic recovery which still possess a high degree of uncertainty despite benefiting from the gradual recovery in consumption and steady growth in exports of services. However, private investment remains concentrated in some business sectors such as retail sale and telecommunications. Nevertheless, PPP will partly support private investment starting from the end of 2016 onwards.

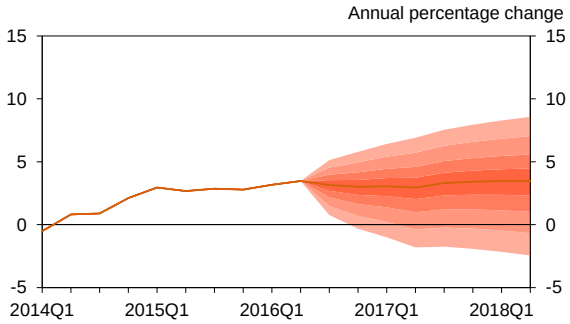
In summary, Thailand's economic growth in 2016 has improved compared with the previous assessment on account of better-than-expected growth outturns in the second quarter, especially in private consumption. However, growth is held back by the lower-than-expected recovery of trading partners' economies and tourism that is affected by the bombing incidents in August and the government's attempt to curb zero-dollar tours. **Growth in 2017 is consistent with the previous assessment.** The slower-than-expected trading partners' economic growth is compensated for by continually increased government expenditure, particularly public investment, in line with the expedited disbursement measure and improved policy certainty.

Risks to Growth and Inflation Forecasts

The Committee views risks to growth to remain tilted toward the downside. Over the forecast horizon, the growth fan chart is skewed down more than previously assessed (Chart 1.4). Compared with the previous *Monetary Policy Report*, increased downside risks mainly stem from the impact from Brexit that will restrain the global economic recovery. Other risks to be closely monitored include (1) risks in the Chinese financial sector, (2) the private sector's limited ability to cope with economic shocks, and (3) impacts on tourism from measures to curb zero-dollar tours and bombings in the seven provinces.

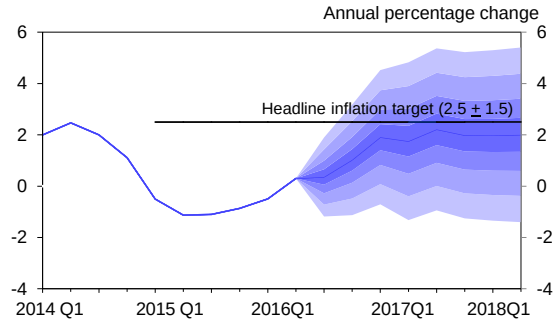
On the contrary, upside risks to the growth forecast are (1) government expenditure in investment projects that may be carried out more quickly and with larger-than-expected outlays, and (2) the positive effect from government measures that may be higher than expected. With regard to inflation, **the Committee judges the balance of risks on both headline and core inflation forecasts to be tilted to the downside in line with the balance of risks to growth (Chart 1.5 and 1.6).**

Chart 1.4 GDP growth forecast



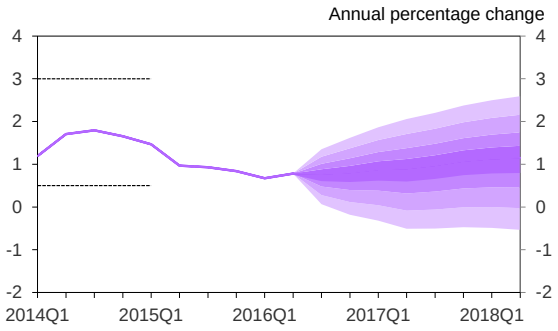
Note: Fan chart covers 90 percent of probability distribution

Chart 1.5 Headline inflation forecast



Note: Fan chart covers 90 percent of probability distribution

Chart 1.6 Core inflation forecast



Note: Fan chart covers 90 percent of probability distribution

1.2 Monetary policy decision

In the third quarter of 2016, monetary policy remained accommodative to support the economic recovery that continued to face greater downside risks on the external front. The Committee emphasized the need to preserve policy space given that external risks still possessed a high degree of uncertainty but could significantly affect the ongoing economic recovery. Hence, the Committee will continue to closely monitor and evaluate risk developments going forward.

At the MPC meeting on August 3, 2016 the Committee voted unanimously to maintain the policy rate at 1.50 percent. The Committee assessed **the Thai economy to continue expanding at a pace close to the assessment in the previous meeting but faces greater downside risks** from heightened uncertainties in the global economy post-Brexit. These included the trade and investment between the UK and the EU, as well as concerns over the European financial sector and political developments abroad. Nonetheless, results obtained from a scenario analysis suggested that different Brexit scenarios can lead to varying economic consequences and their corresponding policy responses. Such exercise will be beneficial to formulating an appropriate policy decision going forward should the situation deviate from the baseline. Meanwhile, **headline inflation was projected to return to the lower bound of the target band within the second half of the year, but there remained risks** from weak demand and lower-than-expected energy prices. Specifically, the timing of the return of headline inflation to the target band would depend mainly on global oil price movements.

Monetary conditions remained eased and conducive to economic recovery. Financing costs remained low as reflected in negative real interest rates. Meanwhile, total corporate financing and household credit continued to expand, although some businesses still faced limitations in obtaining credit amid the gradual economic recovery. **Nevertheless, the baht strengthened over the recent periods, which might not be beneficial to the economic recovery as much as it could.**

The Committee viewed that overall economic conditions in Thailand did not significantly change from the previous meeting, but the economy faced greater downside risks from uncertainties abroad. The decision to keep the policy rate on hold at this meeting was therefore to preserve policy space in order to cushion potential impact that could hinder the ongoing economic recovery. Going forward, the Committee will continue to monitor risks and assess potential impacts as to formulate the appropriate policy response. In addition, the Committee noted that, under the prolonged low interest rate environment, monetary policy must take into account financial stability risks from the search-for-yield behavior that may lead to the underpricing of risks and accumulation of imbalances in the financial system, all of which continued to warrant close monitoring.

At the following meeting on September 14, 2016, the Committee voted unanimously to maintain the policy rate at 1.50 percent. In deliberating their decision, the Committee assessed that **the Thai economy recorded higher-than-expected growth in the second quarter on the back of a strong expansion in private consumption, partly as a result of**

temporary factors including car promotional campaigns and government stimulus measures during the holiday periods. Meanwhile, tourism and public spending remained the key drivers of growth. **Headline inflation rose slightly from an increase in fresh food prices and was expected to return to the lower bound of the target band by the end of this year in line with the previous assessment.** Monetary conditions also remained accommodative and conducive to the economic recovery.

In deliberating their decision, the Committee gave due considerations to the Thai economy which would continue to recover despite the fact that growth momentum may moderate slightly in the second half of the year after a strong acceleration due to temporary factors in earlier periods. In 2017, the economy would continue to expand, as reflected in the closing of the output gap that picked up pace compared with the previous estimate. Concurrently, headline inflation was projected to gradually return to the target band within this year.

However, going forward, there remained risks from the fragile global economic recovery especially from and political uncertainties abroad as well as domestic risks such as the impact of government regulations to curb zero-dollar tours. The Committee therefore emphasized the need to preserve policy space and would continue to closely monitor risk developments.

Overall monetary conditions were accommodative and conducive to economic recovery as real interest rates and bond yields remained low. The recent slowdown in commercial bank credit growth was mainly caused by demand constraints. However, the Committee assessed that **the recent appreciation of the Thai baht**

might not be beneficial for growth and there remained uncertainties that might affect the movement of the Thai baht going forward. In particular, monetary policy of major advanced economies in the low-for-longer environment can contribute to greater capital and exchange rate volatility. Notwithstanding, Thailand's strong external stability would help the economy cope with the increased volatility. However, it can also attract more fund flow that would exert additional pressure on exchange rates.

In addition, the Committee assessed that **overall financial stability remained sound but there remained pockets of risks. These included the search-for-yield behavior under the prolonged low interest rate environment and declining credit quality in some business sectors.** The latter was particularly pertinent to small-and-medium-sized enterprises (SMEs), which financial institutions remained cautious in extending credit to.

Going forward, the Committee affirm that monetary policy should continue to be accommodative. The Committee stand ready to utilize appropriate policy tools so that overall monetary conditions will remain conducive to economic recovery and financial stability.

1.3 Appendix: Summary of assumptions and projections

Table 1.3 Forecast for GDP and assumptions			
Percent	2015*	2016	2017
GDP growth	2.8	3.2	3.2
Domestic demand	2.8	3.0	2.5
Private consumption	2.1	2.7	2.1
Private investment	-2.0	1.1	1.7
Government consumption	2.2	3.5	2.8
Public investment	29.8	9.7	7.5
Exports of goods and services	0.2	1.7	1.0
Imports of goods and services	-0.4	-2.7	2.2
Current account (billion, U.S. dollars)	32.0	40.4	31.8
Value of merchandise exports	-5.6	-2.5	-0.5
Value of merchandise imports	-11.3	-6.6	5.6

Note: *Outturns

Table 1.4 Forecast assumptions			
Annual percentage change	2015*	2016	2017
Dubai oil price (U.S. dollar per barrel)	50.8	41.0	50.0
Non-fuel commodity prices (%YoY)	-17.5	-2.8	3.2
Fresh food prices (%YoY)	-4.3	5.5	1.2
Minimum wage in the Bangkok Metropolitan Region (baht per day)	300	300	300
Government consumption (current price) (%YoY) 1/	4.4	6.0	6.6
Public investment (current price) (%YoY) 1/	25.7	9.4	11.4
Fed Funds rate (% at year-end)	0.38	0.63	1.13
Trading partners' economic growth (%YoY) 2/	3.2	2.9	3.1
Regional currencies vis-à-vis the U.S. dollar (Index)3/	150.7	153.3	156.3

Note: 1/ Including spending on water management plans and infrastructure investment projects
2/ Weighted by each trading partner's share in Thailand's total exports
3/ Appreciation against the US dollar indicated by the minus sign
* Outturns

Table 1.5 GDP growth forecasts by research houses		
	2016	2017
Maybank Kim Eng	4.0	5.0
Thanachart Securities	3.5	4.8
KGI Securities	3.4	3.7
TISCO Securities	3.4	3.6
TMB Bank	3.3	3.5
KT ZMICO	3.3	3.3
JPMorgan	3.3	3.0
FPO [1/]	3.3	-
NESDB [2/]	3.3	-
Bank of Ayudhya	3.2	3.3
Phatra Securities	3.2	3.2
BOT	3.2	3.2
Moody	3.1	3.0
Kasikorn Research	3.0	3.3
Siam Commercial Bank	3.0	3.3
Capital Economic	3.0	2.5
Nomura	2.8	3.0
HSBC	2.8	2.8

Note: Compiled and published by Reuters on September 14, 2016, except:
[1/] Published on July 28, 2016
[2/] Published on August 15, 2016 with the release of GDP data for 2016 Q2
Presented in descending order of 2016 forecasts

Table 1.6 Headline inflation forecasts by research houses		
	2016	2017
Maybank Kim Eng	2.0	2.5
Thanachart Securities	1.7	2.0
Kasikorn Research	0.6	1.8
Nomura Co Ltd	0.6	1.4
FPO [1/]	0.6	-
TMB Bank	0.5	2.0
Capital Economic	0.5	2.0
Siam Commercial Bank	0.4	2.6
TISCO Securities	0.4	2.2
NESDB [2/]	0.4	-
BOT	0.3	2.0
HSBC	0.3	2.0
Moody	0.3	1.8
Bank Ayudhya	0.3	1.7
KGI Securities	0.3	-
KT ZMICO	0.2	1.6
JPMorgan	0.1	1.6

Note: Compiled and published by Reuters on September 14, 2016, except:
[1/] Published on July 28, 2016
[2/] Published on August 15, 2016 with the release of GDP data for 2016 Q2
Presented in descending order of 2016 forecasts

Table 1.7 Probability distribution of GDP growth forecast								
Percent	2016		2017				2018	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
10-12	0	0	0	0	0	1	1	2
8-10	0	0	1	2	3	4	5	5
6-8	1	4	6	8	10	12	12	13
4-6	21	22	22	21	22	22	21	21
2-4	54	40	34	29	27	26	24	23
0-2	22	26	25	24	21	20	19	19
(-2)-0	2	7	10	12	11	11	11	11
< (-2)	0	1	2	5	5	5	6	7

Table 1.8 Probability distribution of headline inflation forecast								
Percent	2016		2017				2018	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
> 7	0	0	0	0	0	0	0	1
6-7	0	0	0	1	1	1	1	1
5-6	0	0	1	2	4	3	4	4
4-5	0	1	6	6	10	8	8	9
3-4	0	4	15	13	16	15	15	15
2-3	3	15	24	20	21	20	19	19
1-2	19	29	25	22	20	20	20	19
0-1	41	29	18	18	14	16	16	15
(-1)-(0)	29	16	9	11	8	10	10	10
(-2)-(-1)	7	5	3	5	3	5	5	5
(-3)-(-2)	1	1	1	2	1	2	2	2
(-4)-(-3)	0	0	0	0	0	0	1	1
< (-4)	0	0	0	0	0	0	0	0

Table 1.9 Probability distribution of core inflation forecast								
Percent	2016		2017				2018	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
> 4.5	0	0	0	0	0	0	0	1
4.0-4.5	0	0	0	0	0	0	0	0
3.5-4.0	0	0	0	0	0	0	0	0
3.0-3.5	0	0	0	0	0	1	1	1
2.5-3.0	0	0	0	1	2	3	4	4
2.0-2.5	0	1	3	4	6	8	9	10
1.5-2.0	2	7	11	13	14	16	16	16
1.0-1.5	22	24	24	22	22	22	21	20
0.5-1.0	47	35	29	25	23	21	20	19
0.0-0.5	25	24	20	19	17	16	15	14
(-1)-0.0	4	8	9	11	10	9	8	8
(-2)-(-1)	0	2	3	4	4	4	4	4
< -2	0	0	1	1	1	1	1	1

Recent Economic Developments

2. Recent Economic Developments

The global economy in the second quarter grew at a slower pace due to a slowdown in major economies. Going forward, the slowdown in global growth was expected to continue given Brexit. The euro area economies were directly affected through trade and private sector's sentiment, while the US and Japanese economies were affected mainly through exchange rates. Asian economies were indirectly affected through exports to the euro area and financial market volatility. Nonetheless, Asian economies were expected to slowly pick up thanks to fiscal measures and an improved export outlook that benefited from the technology cycle with the launch of new product models. Meanwhile, China's growth continued to slow down as a result of ongoing economic reforms and measures to foster macroeconomic stability.

The Thai economy expanded at a gradual pace on the back of public spending and tourism that continued to grow despite having slowed down from the previous quarter. Private consumption steadily expanded due to improved employment in the service sector, the gradual recovery in farm income, and partly a result of temporary factors. Meanwhile, exports continued to contract owing to a slowdown in Thailand's trading partners especially Asian economies, and thus overall investment remained low.

Headline inflation fell slightly on account of fresh food and energy prices, while core inflation remained stable due to low demand pressures. Looking ahead, headline inflation was expected to return to the lower bound of the target band within this year in line with the previous assessment, although the timing would largely depend on developments in global oil prices.

2.1 The global economy

Advanced Economies

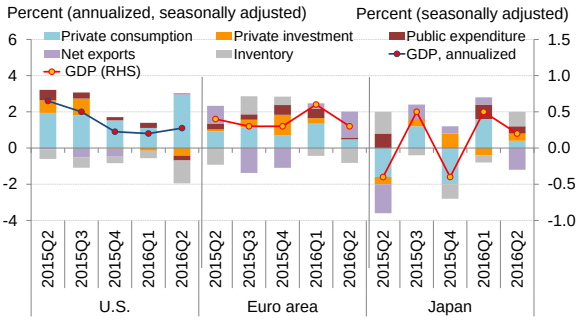
Advanced economies gradually recovered driven by private consumption but faced greater risks following Brexit **(Chart 2.1).**

The result of the referendum held on June 23, 2016, for the United Kingdom (UK) to leave the European Union (EU) (Brexit) dampened economic growth in the UK through declines in business investment and confidence. The impact of Brexit on UK economic growth was expected to increase, particularly in 2017. *(BOX: The impact of Brexit on Global and Thai economies)*

Euro area economies in the second quarter of 2016 grew by 0.3 percent (qoq sa), down from 0.5 percent in the previous quarter on account of domestic demand[1/]. This was mainly attributed to the slowdown in investment, which was partly due to an acceleration in construction work in the previous quarter when the weather was unusually warm, together with the closure of oil refineries in France after worker strikes in May. Meanwhile, net exports improved. Recent economic indicators in August showed more apparent impact from Brexit, as reflected in lower consumer confidence in the euro area and business confidence in Germany.

Looking ahead, the euro area economies were expected to recover slowly with increased risks from (1) the adverse impact from Brexit on the euro area through trade and uncertainties surrounding the new trade deals between UK and the EU which would

Chart 2.1 Source of growth of G3 economies
(percent change from last quarter)



Source: Bureau of Economic Analysis, Eurostat, Cabinet Office of Japan

[1/] Economic data in this Monetary Policy Report are as of September 13, 2016, one day prior to the latest MPC meeting.

continue to affect businesses' and investors' confidence in financial markets; (2) greater political uncertainty that could stem from elections in many countries, especially in France, Italy, and the Netherlands, which could allow anti-EU groups to gain more support; and (3) risks in the European banking sector following the negative interest rate policy that further depressed net interest margin (NIM) in the face of lower credit demand amid greater economic risks. The Committee would continue to closely monitor these risks going forward.

The U.S. economy in the second quarter of 2016 recovered gradually and expanded by 1.1 percent (qoq saar), up from 0.8 percent in the previous quarter on the back of a steady expansion in private consumption. However, investment was held back, especially in energy-related and manufacturing sectors, as a result of depressed oil and commodity prices, the appreciation of the US dollar, and weak external demand. Moreover, inventories declined for three consecutive quarters due to decline in capital goods as investment contracted. Latest economic indicators reflected a slight improvement in many manufacturing industries. Private consumption expanded on the back of continually increasing employment and income. However, exports remained weak in line with trading partners' economic conditions.

In the period ahead, the US economic recovery would likely proceed at a gradual pace, spurred by consumption as supported by the labor market recovery. A continuing recovery in consumption would benefit production in some industries, but weak investment would likely persist particularly in the energy sector. Moreover, uncertainties might arise from (1) the presidential election in November 2016

and its implications for economic policies, (2) external factors such as the impact of Brexit that could induce US dollar appreciation, and weakened demand in the UK and the euro area. However, the continued economic recovery especially with improvements in the labor market and the gradual rise in inflation strengthened the case for the Federal Reserve to raise the federal funds rate in the period ahead.

The Japanese economy grew by 0.2 percent (qoq sa), down from 0.5 percent in the previous quarter. Despite the government's accelerated spending on repair and reconstruction after the earthquake, consumption and exports remained weak as confidence and demand from trading partners continued to decline. Recent economic indicators suggested some improvements in the manufacturing sector as outstanding purchase orders picked up. However, consumption was still weak overall as both wage growth and consumer confidence remained low.

Looking ahead, the Japanese economy would gradually expand, supported by government stimulus measures such as supplementary budgets, a delay in the implementation of the sales tax increase from April 2017 to October 2019, as well as continued monetary policy easing. However, major risks remained from (1) volatility in the global financial market after Brexit, (2) potential yen appreciation, and (3) the transmission of monetary and fiscal policy to the economy which might be less than previously expected.

Asia (excluding China and Japan)

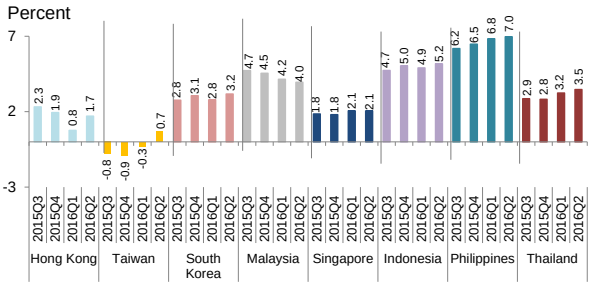
Asian economies recovered at a gradual pace on the back of government expenditure, the initial recovery in exports, and continued expansion in in consumption. However, growth projections for the

Asian economies over the period ahead was lower than the assessment in the previous *Monetary Policy Report*, mainly due to Brexit.

Asian economies (excluding China and Japan) improved in the second quarter of 2016 **(Chart 2.2)** on account of consumption and public investment, together with improved exports in many countries, such as Hong Kong, Taiwan, South Korea, Malaysia, and Singapore, as new products in the technology cycle were to be launched in the third quarter. Private consumption continued to expand. Meanwhile, private investment slowed down due to overcapacity in the manufacturing sector and weakened confidence stemming from the slow economic recovery of trading partners. Latest indicators pointed to a slight slowdown which could be attributed to temporary factors namely the Ramadan which accelerated the production and delivery of merchandises in the second quarter and deferred car purchase in anticipation of newer models at the end of the third quarter.

In the period ahead, Asian economies (excluding China and Japan) would recover gradually following growth-accommodating fiscal policy, gradual improvement in exports, and continued expansion in private consumption. However, the recovery of Asian economies in the period ahead might be weaker than previously assessed in the last *Monetary Policy Report*, due mainly to Brexit. This was particularly the case for Hong Kong, Singapore, Malaysia, and Taiwan, given their closer linkage to the UK and the euro area through trade, direct investment, and financial channels.

Chart 2.2 GDP growth of Asian economies
(percent change from same quarter last year)



Source: CEIC

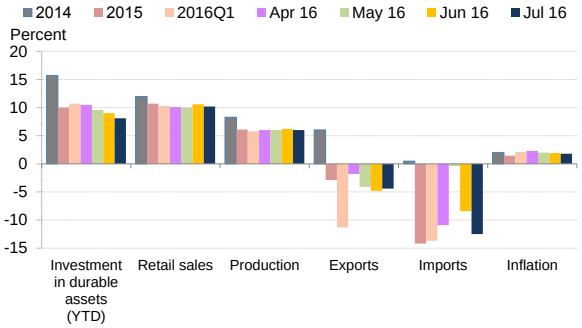
China

China's economic growth continued to moderate, particularly investment growth that was affected by ongoing economic reforms and measures to foster macroeconomic stability.

The Chinese economy in the second quarter of 2016 expanded at 6.7 percent as in the previous quarter. The slowdown in private consumption was compensated by improvements in exports and public investment in infrastructure. However, recent economic indicators **(Chart 2.3)** still pointed to a slowdown in economic growth, particularly investment in the manufacturing and housing sectors. This was due to the ongoing economic reforms to decrease investment in sectors with high degrees of overcapacity, together with measures to foster macroeconomic stability that aimed to curb property speculation and debt accumulation, dampening activities in the real estate sector.

Looking ahead, China's economic slowdown would continue further as a consequence of ongoing economic reforms. The impact of Brexit on the Chinese economy however appeared to be limited, as the share of Chinese exports to the UK and the euro area accounted for only 3.3 percent of GDP. Meanwhile, the impact of Brexit through direct investment and the financial sector would also likely be limited. Financial stability risks in China remained largely unchanged but continued to warrant close monitoring, namely the high levels of corporate debt, and defaults in bank loans, bond markets, and shadow banking. These risks in part prompted the Chinese authority to be more cautious in easing monetary policy further.

Chart 2.3 China's economic indicators
(percent change from same period last year)



Source: CEIC

2.2 The domestic economy

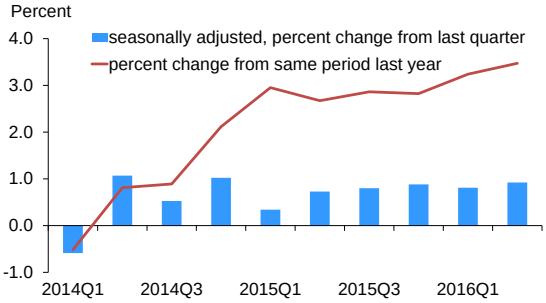
The Thai economy in the second quarter of 2016 continued to expand from the previous quarter **(Chart 2.4)** on the back of a steady growth in public spending and tourism, although both drivers slowed down slightly from the previous quarter. Private consumption continued to expand on account of employment in the service sector, the moderate recovery in farm income, and, in part, temporary factors. The export sector, however, continued to contract in tandem with the slowdown in major trading partners' economies, especially in Asia. Consequently, overall investment lingered remained subdued.

Public spending and tourism continued to expand, although both drivers slowed down slightly after having accelerated in the previous quarter.

Disbursement of the fiscal budget remained on track for both current and capital expenditure, particularly, the expedited disbursement by the National Health Security Fund in the previous quarter and investment on road transportation. However, spending showed some signs of a slowdown after an expedited disbursement in earlier periods **(Chart 2.5)**. Meanwhile, capital expenditure of state-owned enterprises continued to be well disbursed, especially for major projects in the rail infrastructure development plan. In addition, disbursement was on track for the measures to improve living standards in provincial sub-districts.

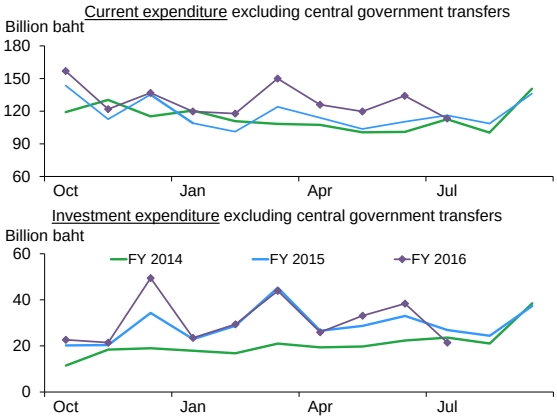
Tourism continued to expand but at a slower rate as a result of the decline in the number of Chinese tourists after having accelerated in the previous quarter **(Chart 2.6)**. The number of tourists from other

Chart 2.4 GDP growth[1/]



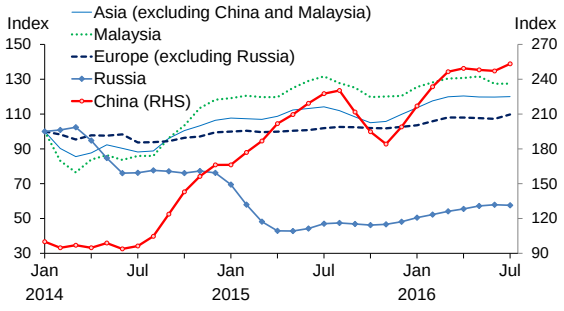
Note: [1/] Calculation based on chain volume measure (CVM)
Source: Office of the National Economic and Social Development Board, seasonally adjusted by Bank of Thailand

Chart 2.5 Public spending



Source: Bureau of Budget; Fiscal Policy Office

Chart 2.6 Index of foreign tourists classified by nationality
(3-month moving average, seasonally adjusted; January 2013=100)



Source: Department of Tourism

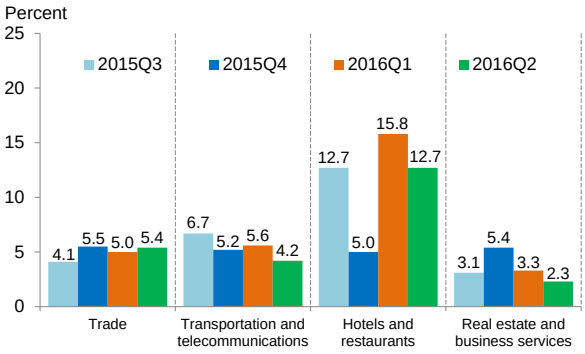
countries steadily recovered. Looking ahead, tourism was expected to expand despite some downside risks, namely (1) the government's attempt to curb zero-dollar tours and measures to set minimum prices for tour packages, and (2) concerns of some tourists over bombings in the seven provinces in August, although the impact might prove to be short-lived as in the past.

The service sector sustained a robust growth in all industries (**Chart 2.7**), especially for tourism-related businesses such as trade, hotels and restaurants, and transport of passengers. These industries all benefited from the higher number of tourists. Meanwhile, the real estate and telecommunication sectors displayed a marginal slowdown after the end of government stimulus measures on the real estate sector and promotional campaigns during the launch of 4G services. The shipping industry grew, albeit at a slower rate as exports remained subdued. On the contrary, the service sector continued to be a major economic driving force and also helped absorb labor that had moved out of the agricultural sector during the drought **(Chart 2.8)**, especially those with low skills.

Private consumption expanded on the back of employment in the service sector and a gradual recovery of farm income and partly because of transitory factors.

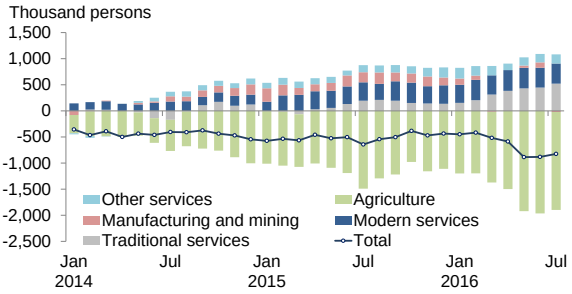
Private consumption recorded stronger growth **(Chart 2.9)**, with expansion seen in almost all components, driven by employment in the service sector and improved farm income after agricultural prices picked up as the drought subsided. Moreover, consumption accelerated partly due to temporary factors, especially spending on durable goods that benefited from the launch of new car models and promotional campaigns, together with the government's

Chart 2.7 Growth of services industries
(percent change from same quarter last year)



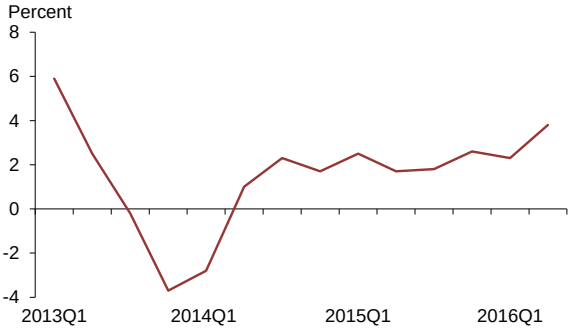
Source: Office of the National Economic and Social Development Board

Chart 2.8 Change in employment from July 2013
(3-month moving average, seasonally adjusted)



Note: Total employment in 2015 was 38 million persons.
Traditional services include wholesale and retail trade, hotels and restaurants, and construction.
Modern services include transportation and telecommunications, finance, education and real estate.
Other services include public administration, other community and personal services, and private households with employed persons.
Source: Labor Force Surveys, National Statistics Office calculations by Bank of Thailand

Chart 2.9 Growth of private consumption
(percent change from same quarter last year)



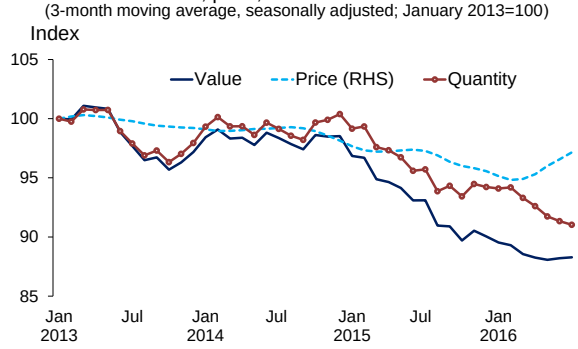
Source: Office of the National Economic and Social Development Board

real-estate stimulus measures that ended in April 2016. In addition, the hotel and restaurant sector gained from the tax rebate scheme during the Songkran holiday. Looking ahead, private consumption would pick up slowly as effects of temporary growth factors and government measures waned.

The value of merchandise exports continued to contract on account of a slowdown in trading partners' economies and increasingly adverse shifts in trade.

Exports of many products continued to contract. The export volume suffered from both a cyclical slowdown in trading partners' economies, especially in Asia, heightened structural changes in global trade as import dependencies declined, and factors specific to Thailand's exported goods **(Chart 2.10)**. Exports of hard disk drives and certain electronic products contracted due to declining global demand as consumers' preference shifted toward tablets and smartphones. Exports of petroleum, petrochemical and chemical products continued to decline as global crude oil prices remained low despite some correction. Exports of agricultural products contracted on account of exports of cassava and rubber to China after, respectively, the Chinese government changed its policy on raw materials for ethanol production and adjusted the import tax on compound rubber. Moreover, agricultural output declined following the drought. Exports of automobile and auto parts expanded well in the second quarter of 2016 as a result of the low base level in the previous year, as Thai manufacturers adjusted production lines for a pick-up truck models in the second quarter of 2015. Going forward however, export of automobile and parts might contract due to a slow recovery in global demand, especially given lower

Chart 2.10 Thai exports (excluding gold)
value, price, and volume indices
(3-month moving average, seasonally adjusted; January 2013=100)



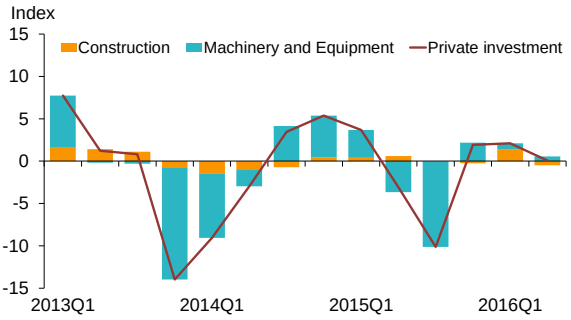
Source: Customs Department, Ministry of Commerce
calculations by Bank of Thailand

purchasing power in the Middle East that resulted from low oil prices.

Private Investment remained subdued following a slow economic recovery.

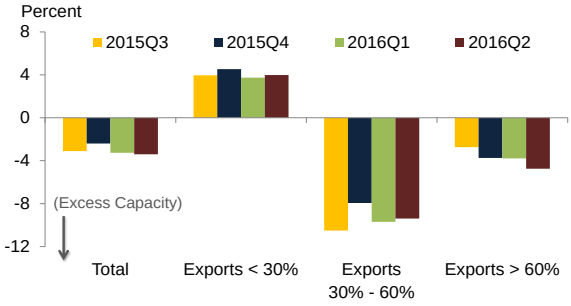
Private investment remained low (Chart 2.11) as exports of merchandise remained subdued. As a result, businesses still had excess capacity **(Chart 2.12)**, lacked demand for new investment. This led to a contraction in imports of capital goods, especially machinery and equipment, together with a slightly lower demand for business credit for investment purposes. Moreover, permitted areas for constructions of housing contracted again after the end of real estate stimulus measures. Nonetheless, the marginal increase in investment was attributable to increased sales of domestic machineries, especially in the plastics industry, and sales of commercial vehicles. In addition, investment in some service sectors continued to grow, for example, in machineries used in power generators, telecommunications equipment, and smartphones.

Chart 2.11 Contribution to growth of private investment
(percent change from same quarter last year)



Source: Office of the National Economic and Social Development Board

Chart 2.12 Capacity utilization

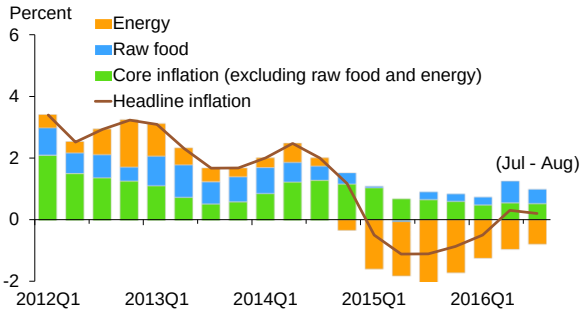


Note: Calculated from the difference between capacity utilization in each quarter and 2011 - 2013 average, excluding period of severe flooding (October 2011 - January 2012)
Source: Calculations by Bank of Thailand

2.3 Production cost and price conditions

Headline inflation declined slightly, mainly on account of fresh food and energy prices. Core inflation, on the other hand, remained stable due to a low demand-pull pressure. In the period ahead, the Committee assessed that headline inflation would return to the lower bound of the target band within this year as previously expected, although the timing would depend largely upon developments of global crude oil prices.

Chart 2.13 Contribution to headline inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce calculations by Bank of Thailand

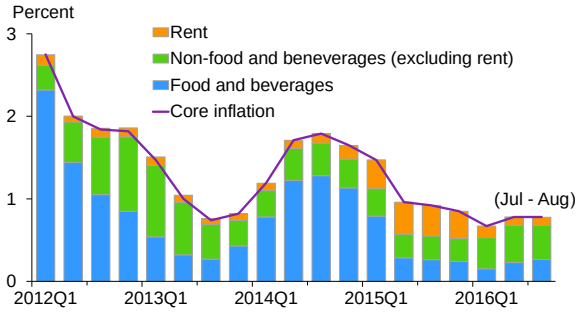
Headline inflation softened slightly, averaging 0.20 percent (yoy) in the first two months of the third quarter **(Chart 2.13)**. This was due to falling prices of fresh food and retail oil. The decline in fresh food prices was due to increased supply of vegetables and fruits produce after the drought subsided. Domestic retail oil prices fell in line with global crude oil prices, as oil supply from the US and OPEC remained elevated, while the global economic recovery faced additional risks.

Core inflation averaged at 0.78 percent (yoy) in the first two months of the third quarter **(Chart 2.14)**, close to the previous quarter. This was due to weak demand-pull inflationary pressure following the gradual recovery of domestic economy, as reflected by underlying inflation indicators that remained at low levels **(Chart 2.15)**. Moreover, prices of most goods and services in the core inflation basket remained unchanged, indicating limitations faced by businesses in making price adjustments.

Short-term inflation expectations stood close to that in the previous period. One-year-ahead inflation expectation by firms in August 2016 and by professional economists in September 2016 stood at 1.9 percent and 1.6 percent, respectively. Medium-term inflation expectations remained near the inflation target at 2.5 percent **(Chart 2.16)**.

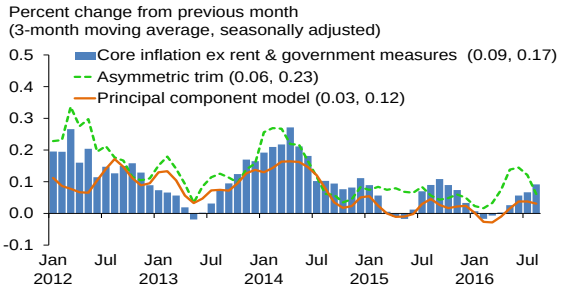
Looking ahead, the Committee expected headline inflation to rise gradually due to the base effect of low oil prices. Headline inflation was also expected to return to the lower bound of the target band within this year as previously assessed. Nonetheless, the timing would largely depend on developments in global crude prices. In addition, the recovery of domestic demand continued to warrant close monitoring.

Chart 2.14 Contribution to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce calculations by Bank of Thailand

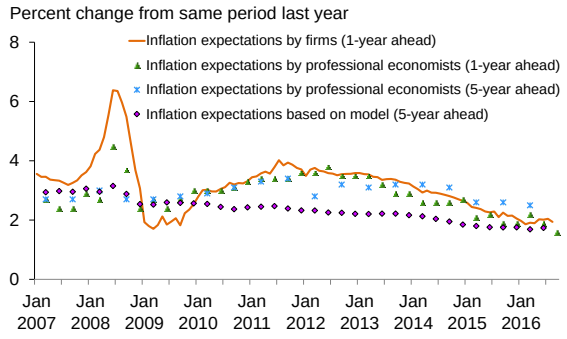
Chart 2.15 Underlying inflation indicators



Note: Data point indicated in () where the first value is %MoM (sa, 3mma) as of August 2016, while the second value is 2004-2014 average; **Asymmetric trim** excludes goods and services with most volatile price changes, removing the bottom 10 percentile and the top 6 percentile; **Principal component model** calculates changes in common statistical components that attribute price movements across categories of goods and services.

Source: Bureau of Trade and Economic Indices, Ministry of Commerce calculations by Bank of Thailand

Chart 2.16 Inflation expectations



Source: Business Sentiment Survey of Bank of Thailand, Consensus Forecast, and calculations based on macro-finance term structure model

Table 2.1 Quarterly inflation								
Unit: Percent	2014	2015				2016		
		Q1	Q2	Q3	Q4	Q1	Q2	Jul-Aug
Percentage change from previous year (%Δyoy)								
- Headline Consumer Price Index (Headline CPI)	1.89	-0.50	-1.12	-1.11	-0.86	-0.50	0.30	0.20
• Core Consumer Price Index (Core CPI)	1.59	1.47	0.96	0.92	0.85	0.67	0.79	0.78
• Raw food	3.46	0.30	-0.40	1.48	1.45	1.52	4.24	2.86
• Energy	1.68	-12.95	-14.16	-16.42	-14.63	-11.41	-8.95	-7.98
Percentage change from previous quarter (%Δqoq_sa)								
- Headline Consumer Price Index (Headline CPI)	-	-0.7	-0.2	0.0	0.0	-0.3	0.6	-
• Core Consumer Price Index (Core CPI)	-	0.3	0.1	0.3	0.2	0.1	0.2	-
• Raw food	-	0.2	-0.5	1.4	0.4	0.2	2.2	-
• Energy	-	-7.4	-2.1	-3.7	-2.4	-3.7	0.46	-

Source: Bureau of Trade and Economic Indices, Ministry of Commerce
Calculations of percentage change from last quarter, seasonally adjusted, by Bank of Thailand

The impact of Brexit on global and Thai economies

Brexit, which resulted in heighted uncertainties in the global economy and global financial markets, was an important factor underpinning a downward revision of global and Thai economic growth forecasts from the projection published in the previous *Monetary Policy Report*.

Most analysts expected that Brexit would set back the UK economy in the period ahead. Domestic spending would slow down due to weakened business and consumer confidence. Moreover, foreign direct investment in the UK might decline significantly given that the UK was the financial center of the European Union (EU), and was highly dependent on capital from the EU. The impact on trade, however, might not be apparent in the short run as a weaker pound could benefit exports and tourism. This is because the EU was a major export destination for the UK, accounting for 39 percent of the UK's total exports or 8 percent of GDP. Nonetheless, the medium- and longer-term impacts would depend on the terms of negotiations with the EU, while the outcomes could have implications on the unity of the EU in the future.

Global economic impact of Brexit

Brexit could impact the global economy through three main channels as follows.

(1) Trade channel

The UK's economic slowdown could dampen global trade. In particular, the effect would come from the **UK's trade with the EU** due to their close economic relationships. The EU's exports to the UK accounted for 6.5 percent of total EU exports or 2.4 percent of GDP. If the impact of Brexit on the EU economy were to be limited, other countries would also be less affected due to their low trade dependence with the UK **(Chart 1)**. The impact of Brexit on Asian economies were assessed to be not significant because each country's exports to the UK accounted for less than 3 percent of their total exports. Singapore and Hong Kong, as the financial centers of Asia, were the exceptions due to their large share of service exports to the UK. However, Asian countries, including Thailand, which had closer trade linkages with the EU could be indirectly affected if the EU economies weakened significantly.

Chart 1 Shares of exports to the UK and EU



Note: ASEAN-5 consists of Singapore, Malaysia, Indonesia, Thailand and the Philippines
Source: CEIC

(2) Confidence channel

The unexpected outcome of the Brexit referendum induced panic among investors and spurred up significant volatility in the financial markets with the risk-off sentiment. As a result, global stock indices declined significantly especially in the advanced economies, and the pound and the euro depreciated, while safe-haven currencies rallied noticeably. Although financial market volatility subsided as the outlook gradually became clearer, risks remained owing to political uncertainty and the stability of regional economic blocs. Within the EU, the intensified

anti-integration sentiment which could give rise to a call for a referendum in many countries similar to the UK. For example, the anti-EU group in Italy gained increasingly greater popularity. **The rising political uncertainty would weigh on investor confidence in the EU economic outlook. Moreover, this could add more pressures on the European banking sector that is now facing with elevated levels of non-performing loans during the economic slowdown since 2008**. In the case of Italy, stock prices in the banking sector fell by more than 15 percent after Brexit. In the period ahead, increased financial stability risks in the banking sector and a growth slowdown in the EU, the world's second largest economy, would affect global trade and investment as well as global economic recovery.

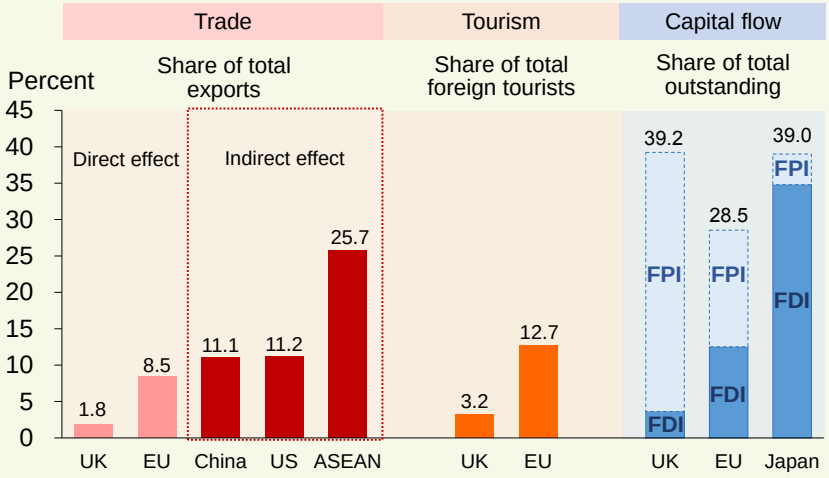
(3) Investment channel

UK's slower economic growth would weigh on returns on investment in the UK, particularly the EU's foreign direct investment (FDI) in the UK which was as high as 12 percent of the EU's total outward FDI. On the other hand, other countries in the region would not be affected much due to their low reliance on the UK's direct investment. However, negotiations on post-Brexit economic relations between the EU and the UK remained highly uncertain, and that could affect investor confidence. With an aim to support the economies, further monetary policy easing by central banks in major advanced economies, especially those affected by Brexit, could prolong the low for longer environment. This could encourage the search-for-yield behavior which would affect capital flows for some time.

The impact of Brexit on the Thai economy

The major impact of Brexit on Thailand would be via the trade channel through the income effect from the slowdown in the EU economies and those with close trade linkages with the EU and the UK **(Chart 2)**. Meanwhile, investment could experience higher volatility, especially in portfolio investment flows.

Chart 2 Linkages between Thai economy and trading partners in various dimensions



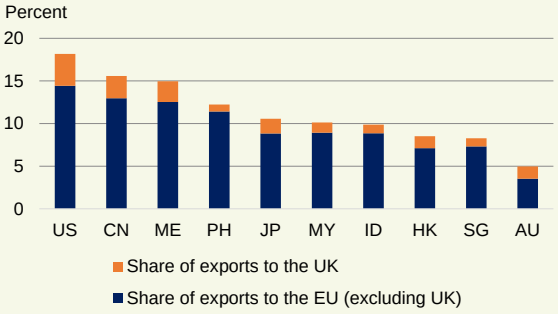
Note: Data in 2015
FPI is foreign portfolio investment

Source: CEIC

(1) Trade channel

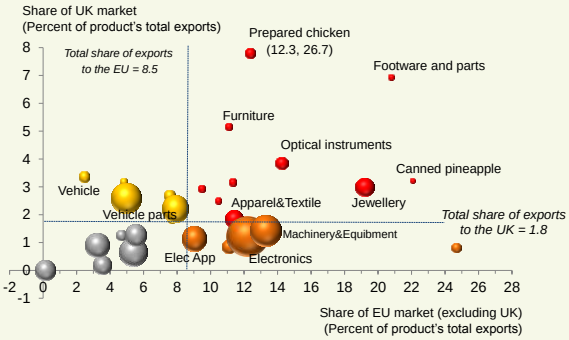
The direct impact of the UK's economic slowdown on Thai exports would be limited as the UK's share in Thai total exports accounted for only 1.8 percent in 2015. However, Thai exports could be **indirectly impacted by the income effect owing to the economic slowdown in (1) the EU** which currently accounted for 8.5 percent of total Thai exports and **(2) other major economies with close trade links with the EU and the UK,** such as the US, China, and ASEAN, which altogether accounted for almost half of total Thai exports **(Chart 3).**

Chart 3 Dependency ratio on the UK and EU (excluding UK) markets of Thailand's major trading partners



Note: Data in 2015
Source: Trademap and the Customs Department, calculations by Bank of Thailand

Chart 4 Thai exported products classified by UK and EU (excluding UK) market dependency



Note: Products in the top-right corner face risks from heavily relying on the UK and EU markets. Size of bubble indicates products' importance in Thai exports (share of Thai total exports)
Source: Customs Department and calculations by Bank of Thailand

Thai exports highly dependent on EU and UK markets (Chart 4) included those with high income elasticity such as jewelries and those with intense price competition such as processed chicken, canned pineapple, and canned fish. These products would suffer from both a slowdown in trading partners' economies and volatile exchange rates. **However, such products accounted for a small share of total Thai exports.** In the future, if the import tariffs of trade between the UK and the EU were to be increased from zero percent, the UK's import demand from the EU could fall. This might bring about trade opportunities for other countries. For instance, Thailand could benefit from exporting more processed food, automotive, textile, and apparels to the UK.

(2) Tourism channel

Brexit could affect Thailand's tourism sector. Accounting for 15.9 percent of total tourists, UK and EU (excluding Russia) travelers are significantly important to Thailand. Brexit could hamper confidence in future income amid an economic slowdown. Also, travelling costs could increase due to the weaker pound and euro. However, the adverse impact might not be seen until 2017 as most Europeans plan their trips on average at least six months in advance.

(3) Investment channel

The Brexit aftermath could affect each type of capital flows differently, with most apparent impact on portfolio investment. After Brexit, Thailand recorded higher-than-average capital inflows in both debt and equity markets, consistent with other emerging markets. This could be attributed to both monetary policy easing in advanced economies and expectations of the Federal Reserve to delay a policy rate hike. Looking ahead, monetary policy

directions in advanced economies would remain a major push factor for investment decisions and consequently impact capital flow volatility.

The impact on FDI was expected to be limited. In the past, Thailand's dependence on FDI from the UK and the EU has been small. Moreover, capital inflows were influenced mostly by the strength of domestic fundamentals acting as a pull factor.

While global financial volatility subsided somewhat after having surged during Brexit, future uncertainty remained high. Implications from Brexit could rekindle volatility in financial markets and the global economy. Therefore, the following developments required close monitoring: (1) the timeframe of the UK withdrawal from the EU, which could take at least two years after the UK's formal notification to the EU, (2) outcome of negotiations on the UK and EU relationship, and (3) repercussions of the Brexit impact on other countries.

Monetary and Financial Stability

3. Monetary and Financial Stability

Volatility in global financial markets surged temporarily following the Brexit referendum and led a number of advanced economies to prolong monetary policy easing further ("low for longer"), which in turn induced investors to increase their holdings of emerging market equity and debt. On the domestic front, monetary conditions remained accommodative, with money market rates remaining at low levels. Meanwhile, corporate costs of financing through the debt market increased slightly from the previous quarter but still remained low overall. With regard to exchange rates, the baht strengthened from both external and domestic factors.

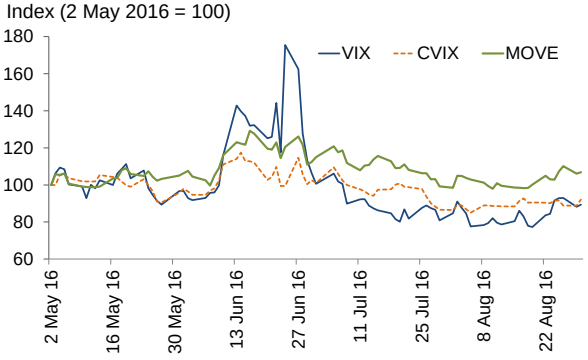
Overall macro-financial stability remained sound. This was reflected in the high levels of capital buffers and loan loss provision maintained by commercial banks to cushion against risks from declining credit quality, as well as the financial standing of large corporates which remained strong. While the search-for-yield behavior remained under the prolonged low interest rate environment, there were still limited signs of risk underpricing. Nevertheless, there remained pockets of risks that warranted monitoring as follows. First, although household debt accumulation had decelerated and improvement in farm income would benefit debt serviceability in the future, already elevated household debt level and uncertainties in the economic recovery would require a certain period of necessary adjustments on part of the households. Second, debt serviceability of the corporate sector, particularly for SMEs, was affected by the slow economic recovery. Third, a slowdown in the condominium market was expected, as housing demand declined after the government's stimulus measures ended.

3.1 Financial markets

Global financial markets

Market volatility heightened temporarily in the third quarter, as indicated by the volatility indices[1/] **(Chart 3.1)**, following the Brexit

Chart 3.1 Volatility indices

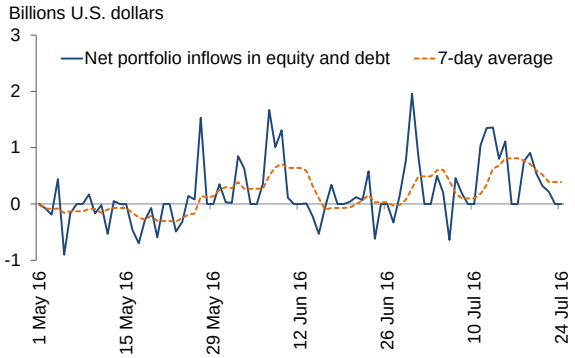


Source: Bloomberg and calculations by Bank of Thailand

[1/] **VIX** (Volatility Index) is an indicator for stock market volatility, measured by implied volatility of option prices of the S&P500.

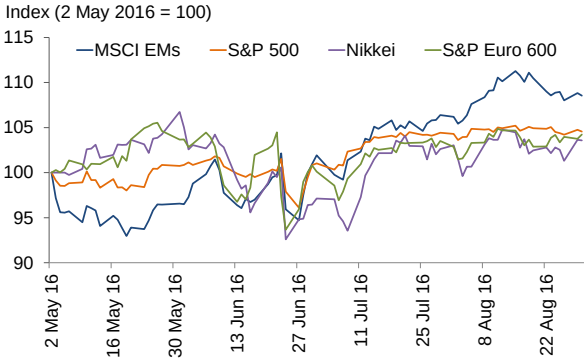
referendum on June 23, 2016. In response, the Bank of England decided to ease its monetary policy further on August 4, 2016. The result of both Brexit and continued monetary policy easing by major central banks, particularly the Bank of Japan and the European Central Bank, led investors to expect that interest rates and bond yields in advanced markets would remain low for longer. This prompted capital flows to return to emerging markets (EMs) in search of higher yields, whereby institutional investors, mutual funds, and exchange traded funds (ETFs) started to increase the weight of EMs in their asset allocation. Consequently, this led to increased capital inflows into EMs, in both equity and bond markets **(Chart 3.2),** which resulted in a greater increase in EMs capital market indices (MSCI EMs[2/]) relative to other markets after Brexit **(Chart 3.3).** However, capital flows into EMs slowed down somewhat at the end of August following comments by some members of the Federal Open Market Committee (FOMC) that alluded the possibility of an increase in the federal funds rate in the remaining meetings in 2016.[3/]

Chart 3.2 Net portfolio flows



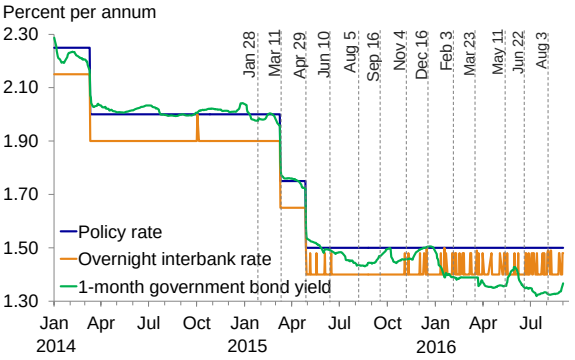
Source: Institutes of International Finance

Chart 3.3 Stock Exchanges



Source: Bloomberg and calculations by Bank of Thailand

Chart 3.4 Money market interest rates



Source: Bank of Thailand and Thai Bond Market Association (Thai BMA)

MOVE (Merrill Lynch Options Volatility Estimate) is an indicator for option bond market volatility, measured by implied volatility of the U.S. government bond market.
CVIX (Currency Volatility Index) is an indicator for foreign exchange market volatility, measured by implied volatility of 9 major currency pairs.

2/ **MSCI EMs** (Morgan Stanley Composite Emerging Markets Index) is a market capitalization index that consists of indices in 23 emerging market economies including Brazil, Mexico, Turkey, China, India, Thailand, Malaysia, and Indonesia.

3/ Economic data in this *Monetary Policy Report* are as of September 13, 2016, one day prior to the latest Monetary Policy Committee meeting.

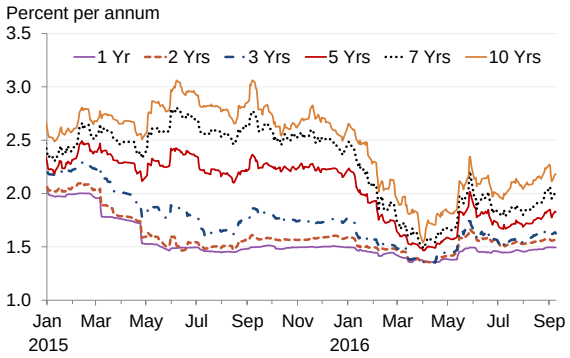
Money and bond markets

Thailand's short-term money market interest rates stayed below the policy rate between July and September 2016 **(Chart 3.4).** Meanwhile, government bond yields remained at low levels, despite edging up slightly from levels seen at the end of the second quarter **(Chart 3.5).** This was driven by improved investor sentiments following the acceptance of the draft constitution in the referendum on August 7, 2016 and the better-than-expected GDP figure for the second quarter of 2016. Investors also expected a possible rate increase by the Federal Reserve by the end of this year after releases of improved US economic indicators.

Credit spreads between corporate and government bond yields were largely stable relative to the previous quarter. As a result, costs of financing through corporate bonds rose slightly along with the increase in government bond yields. However, overall financing costs remained below last year's averages **(Chart 3.6).**

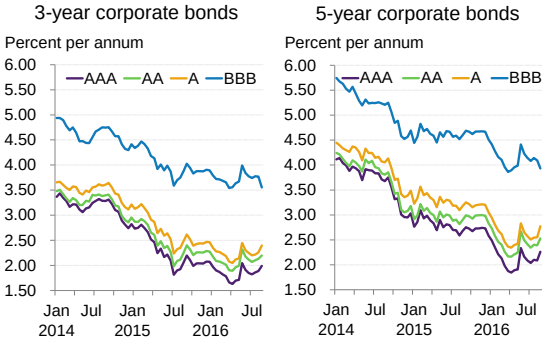
The Stock Exchange of Thailand (SET) Index and the Market for Alternative Investment (mai) Index rose from the previous quarter **(Chart 3.7 and 3.8).** The rebound was attributable to foreign investors who continued to increase investment in the Thai equity markets, partly reflecting the return of capital flows to emerging markets under the low-for-longer environment. This was also supported by improvements in the Thai economy and performances of listed companies during the first half of the year, together with better investor sentiments after the referendum on the draft constitution. However, SET and mai indices sharply fell in early September to levels similar to the end of the second quarter. While local institutional investors and proprietary traders were

Chart 3.5 Thai government bond yields



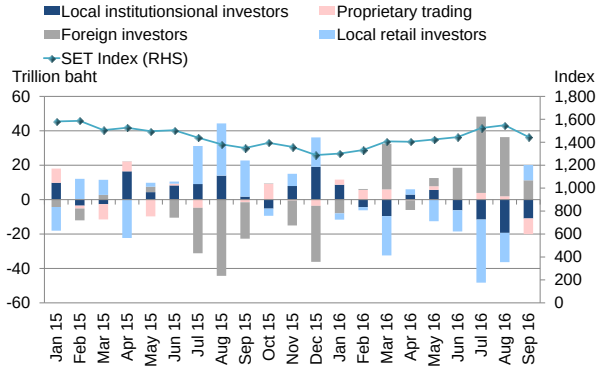
Source: Bank of Thailand and Thai Bond Market Association (Thai BMA)

Chart 3.6 Thai corporate bond yields



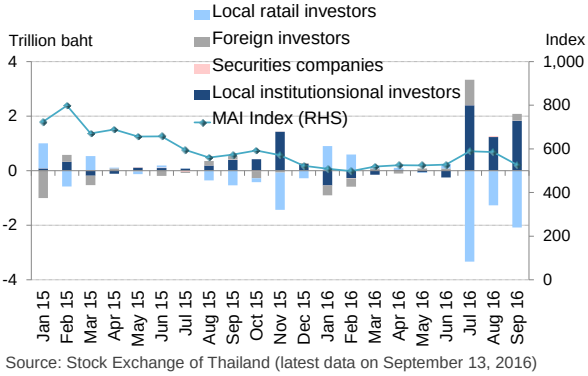
Source: Thai Bond Market Association (Thai BMA)

Chart 3.7 Stock Exchange of Thailand Index and net buy classified by investor types



Source: Stock Exchange of Thailand (latest data on September 13,2016)

Chart 3.8 Market for Alternative Investment (MAI) and net buy classified by investor types



Source: Stock Exchange of Thailand (latest data on September 13, 2016)

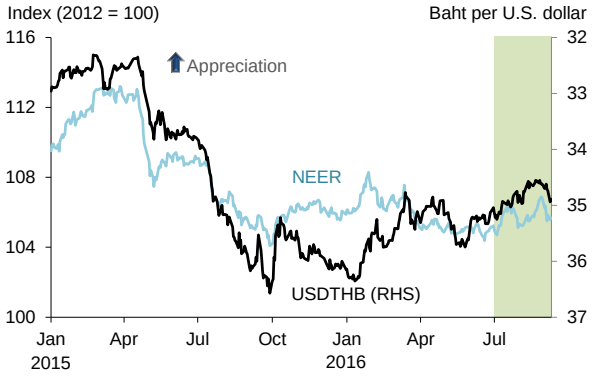
net sellers in SET, retail investors were net sellers in mai.

Foreign exchange markets

The baht appreciated against the US dollar relative to the end of the previous quarter, in line with movements of most regional currencies, as a result of the low-for-longer environment that spurred continued capital inflows into assets in Asia and Thailand. Additional factors driving the baht to strength included the result of the draft constitution referendum and better-than-expected GDP figure for the second quarter. However, the baht and regional currencies depreciated slightly in some periods, partly owing to the appreciation of the US dollar. Factors supporting the US dollar were better-than-expected US economic indicators, such as employment and production figures, and comments by some members of the FOMC that a policy rate hike this year might be appropriate. On September 13, 2016, the baht closed at 34.89 baht per US dollar, up 0.7 percent from the end of the previous quarter. This was in line with the Nominal Effective Exchange Rate which closed at 105.69, up 0.5 percent from the previous quarter, as the baht strengthened against major and most regional currencies **(Chart 3.9 and 3.10).**

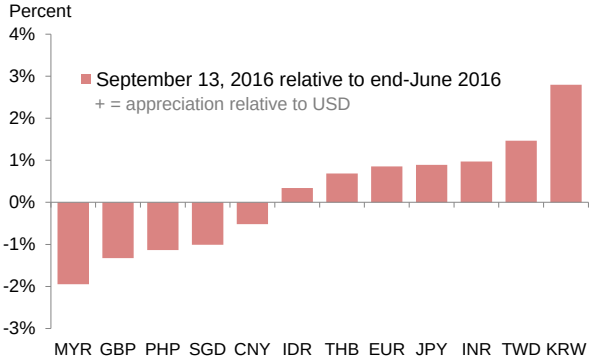
Baht movements were somewhat more volatile in August but relatively stable compared with regional currencies **(Chart 3.11**). Going forward, global capital flows and exchange rates were expected to remain volatile in view of uncertainties in policy directions of major economies and continuing sensitivity of the global financial markets under the low-for-longer environment.

Table 3.9 Movements in USDTHB and NEER



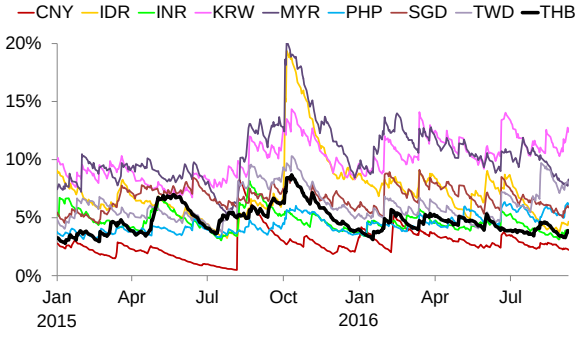
Source: Bank of Thailand and Reuters (latest data on September 13, 2016)

Chart 3.10 Currency movements vis-à-vis USD



Source: Bank of Thailand (latest data on September 13, 2016)

Table 3.11 Volatility of THB and regional currencies



Note: Volatility calculated by Exponentially Weighted Moving Average (EWMA)
Source: Bank of Thailand and Reuters (latest data on September 13, 2016)

3.2 Financial institutions

Costs of financing for firms and deposit rates offered by commercial banks were stable. Although overall demand for new financing rose across the board relative to the first quarter of 2016, demand was concentrated in large corporates and the service sector. Going forward, as the economy was projected to recover only gradually, the outlook for a slowdown in credit growth was expected, thereby reducing the need for deposit mobilization.

Interest rates, credits, and deposits

Commercial bank lending and deposit rates were stable overall as the policy rate remained unchanged **(Table 3.1).**

Table 3.1 Benchmark loans and deposit rates of commercial banks*

Percent per annum	2014	2015		2016		
		Q3	Q4	Q1	Q2	Sep 13
12-month deposits						
Average of the 4 largest commercial banks**	1.73	1.46	1.40	1.40	1.38	1.38
Average of remaining banks***	2.22	1.72	1.59	1.47	1.42	1.41
Minimum lending rate (MLR)						
Average of the 4 largest commercial banks	6.75	6.51	6.51	6.51	6.26	6.26
Average of remaining banks	7.44	7.31	7.25	7.25	7.16	7.16
Minimum retail rate (MRR)						
Average of the 4 largest commercial banks	8.08	7.87	7.87	7.87	7.62	7.62
Average of remaining banks	8.63	8.54	8.54	8.54	8.45	8.45

Note: * Benchmark rates averaged across commercial banks at the end of each period

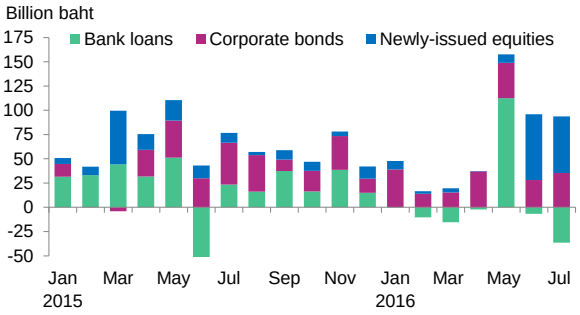
** Four largest commercial banks are BBL, KTB, KBANK and SCB.

*** The following commercial banks are now included under remaining banks: Bank of China (Thailand) (added August 2014), ANZ (Thailand) (added June 2015), and Sumitomo Mitsiu Trust Bank (Thailand) (added October 2015)

Total corporate financing increased between May and July 2016 relative to the first quarter in all channels (Chart 3.12), mainly thanks to increased fund raising by large corporates. Overall commercial bank loan growth accelerated, primarily driven by demand for domestic and offshore merger and acquisition deals. However, excluding loans for these purposes, new loans continued to decline from the first quarter of 2016, especially loans to the manufacturing sector given that private investment remained low. In addition, some large corporates, especially those in the transport and logistics, telecommunication, and food processing sectors, shifted to bond financing as costs of corporate bond issuance remained at relatively low levels. As a result, the amount of funds raised from the bond market between May and July 2016 increased from the first quarter of 2016. Funds raised from equity financing also increased, mainly by telecommunication and trade businesses.

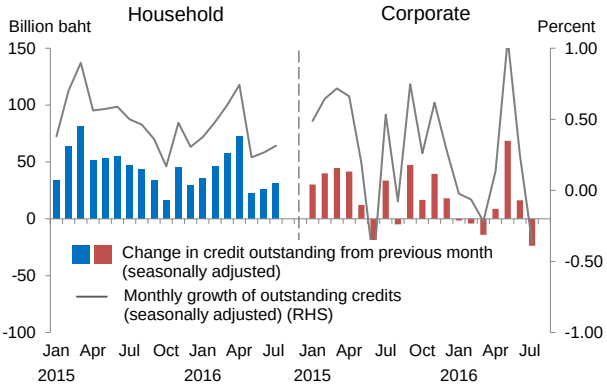
New loans to households[4/] between May and July 2016 increased at a slower rate relative to the first quarter of 2016 (Chart 3.13). The slowdown was seen across the board, noticeably in mortgage loans which had earlier accelerated due to the government's real estate stimulus measures. Meanwhile, new loans from specialized financial institutions and savings cooperatives accelerated somewhat. The slowdown in new loans to households was consistent with the Credit Conditions Survey in the second quarter of 2016. The Survey indicated that financial institutions continued to maintain tight

Chart 3.12 Total corporate financing by instuments*



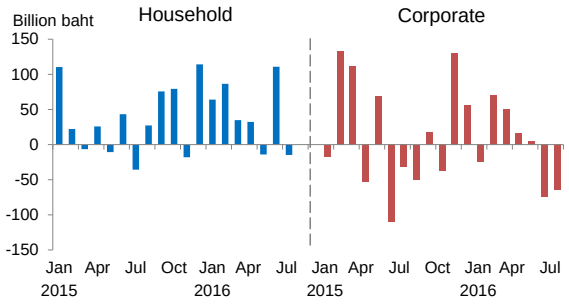
Note: *Monthly change in corporate loans (seasonally adjusted), corporate bonds excluding commercial banks, and newly issued equities.
Source: Bank of Thailand and Thai BMA

Chart 3.13 New private credits



Note: *New corporate credits calculated from seasonally adjusted change in credits outstanding of Other Depository Corporations (ODCs)
Source: Bank of Thailand

Chart 3.14 New deposits*



Note: *Change in outstanding deposits at depository institutions (excluding Bank of Thailand) from the previous month. The amounts only account for new deposits, not including transfers within and between commercial banks or rolled-over deposits.
Source: Bank of Thailand

[4/] New loans are calculated from the changes of seasonally adjusted credit outstanding of other depository corporations (ODCs), which include commercial banks, specialized financial institutions, finance companies, savings cooperatives, and money market funds.

lending standards on households, which might instead turn to non-banks.

Deposits outstanding at the end of July declined from the end of April 2016, especially for corporate deposits (Chart 3.14). Growth of deposits including bills of exchange (B/E) remained low given weak loan growth and the subdued outlook that reduced the need for deposit mobilization. This was reflected in the low level of interest rates on special deposit products **(Chart 3.15).**

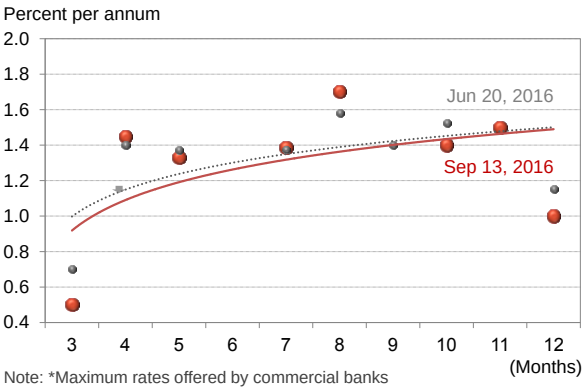
Overall, **the loan-to-deposit (including B/E) ratio rose from the first quarter of 2016** due to increases in new loans and declining deposits **(Chart 3.16).**

Macro-financial stability

Overall macro-financial stability remained sound, despite declining loan quality, particularly in the SME sector, due to sluggish economic recovery. In addition, stability of financial institutions remained sound as commercial banks continued to monitor credit quality closely and maintained high levels of capital buffers and loan loss provisions to provide cushion against risks from declining loan quality. Meanwhile, the search-for-yield behavior continued but systemic risks remained limited.

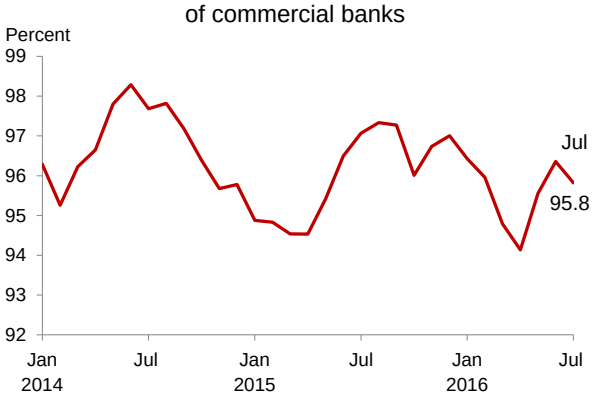
The quality of commercial bank loans at the end of the second quarter of 2016 declined from the previous quarter given the sluggish economic recovery. The ratio of non-performing loans to total loans (NPL ratio) rose from 2.64 at the end of the first quarter of 2016 to 2.74, attributable to the increase in NPL ratios of both business loans, especially loans to SMEs, and consumer loans **(Chart 3.17).** The NPL ratio of SMEs rose from 3.69 at the end of the first quarter of 2016 to

Chart 3.15 Special deposit rates*



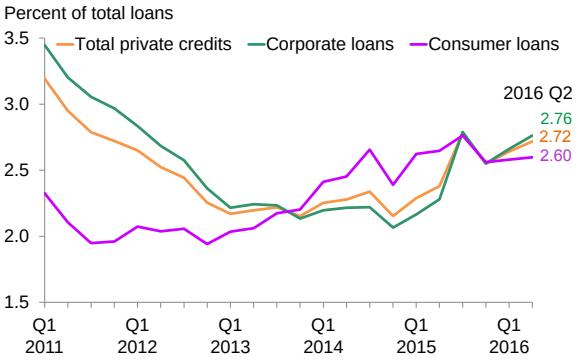
Note: *Maximum rates offered by commercial banks
Source: Bank of Thailand

Chart 3.16 Loan to deposit (including B/E) ratio of commercial banks



Source: Bank of Thailand

Chart 3.17 Non-performing loans of commercial banks



Source: Bank of Thailand

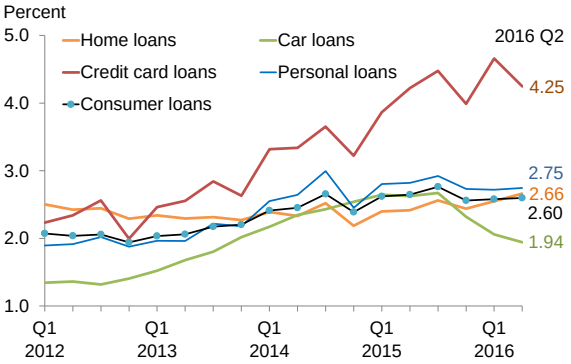
3.77. The NPL increase was due to loans extended to the manufacturing and wholesale and retail trade sectors. The NPL ratio of loans to large corporates also increased but remained low at 1.85, while the NPL ratio of consumer loans edged up slightly due to declining quality of housing and personal loans **(Chart 3.18).**

Despite declining loan quality, stability of overall commercial banks remained sound as banks maintained high levels of capital buffers and loan loss provisions, as reflected in the actual-to-regulatory loan loss provision ratio at 161.3 percent, together with the capital adequacy ratio at 17.5 percent that was well above the regulatory requirement of 8.5 percent.

The search-for-yield behavior continued under the prolonged low interest rate environment. With regard to equity market movements, the turnover ratio in the SET and mai increased from the previous quarter while the SET Index picked up following the acceptance of the draft constitution in the referendum and improved performance of listed companies in the second quarter. However, in early September, the SET Index was down sharply owing to sales by domestic proprietary traders and institutional investors. **As a result, the price-earnings ratios (P/E) for the SET and mai declined below the levels seen at the end of the second quarter but remained higher than historical averages (Chart 3.19).** However, **risks to the overall stability of the financial system were expected to be limited** as transactions were mainly done in cash, while margin loans remained low at 13 percent of total transaction value as of July 2016.

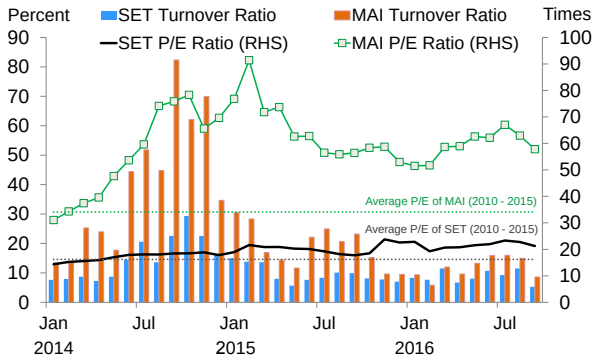
With regard to the domestic bond markets, corporate bond issuance continued to rise over time. Overall, most corporate bond

Chart 3.18 Share of non-performing loans in consumer loans



Source: Bank of Thailand

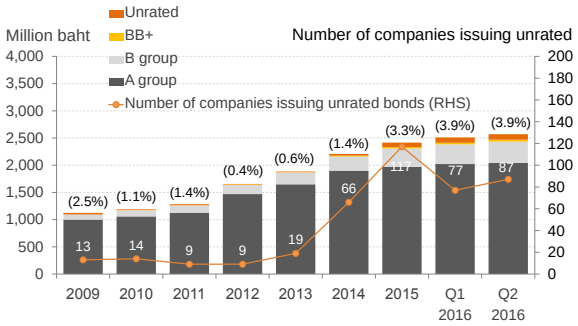
Chart 3.19 Turnover ratio and Price-Earnings Ratio (P/E)* in SET and MAI



Note: *Current price to Earnings Per Share (EPS) over the last 12 months
Source: Stock Exchange of Thailand (latest data as of September 13, 2016)

outstanding possessed good credit ratings **(Chart 3.20). Nonetheless, unrated bond issuance were also on the rise, a portion of which were issued by non-listed companies that had higher risks** as reflected by the debt-to-equity (D/E) ratios that were higher than the averages of listed companies in each corresponding sector. These unrated issuers also had annualized interest coverage ratios (ICR) that were on average lower than those of listed companies **(Table 3.2).** However, systemic risks were still limited as unrated bonds constituted only a small share of total corporate bonds outstanding, at 3.9 percent, and were mainly sold to accredited investors.

Chart 3.20 Corporate bonds outstanding
(short and long maturities) classified by credit rating



Note: The A group consists of AAA to A-, while the B group consists of BBB+, BBB and BBB-
() Percent of non-rated bonds in total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

Table 3.2 Financial position of unrated-bond issuing companies
(classified by listed and non-lited companies)

Sector of unrated-bond issuing companies	Outstanding at Q2/16 (million baht)	D/E (times)		D/E in SET**	ICR (times)		ICR in SET**
		Listed	Non-listed*		Listed	Non-listed*	
Real estate	31,252 (40.6%)	1.4	2.7	1.3	3.9	2.0	6.5
Energy	23,035 (29.9%)	1.2	1.4	1.2	2.0	0.4	2.7
Professional services	3,157 (4.1%)	0.6	1.7	0.4	3.2	0.2	11.2
Tourism	2,340 (3.0%)	1.0	2.8	0.9	0.9	0.6	4.8
Other	17,226 (22.4%)	1.0	1.8	0.7	5.6	0.9	5.6

Note: ( ) displays a share of unrated bond within sector
D/E and annualized ICR is a weighted average
Listed refers to companies listed in SET and MAI
Non-listed refers to companies not listed in a stock exchange
* Data from Department of Business Development (BOL Database) in 2014
** Data are weighted averages of all companies in each sector of SET.
Source: ThaiBMA, Department of Business Development and Stock Exchange of Thailand

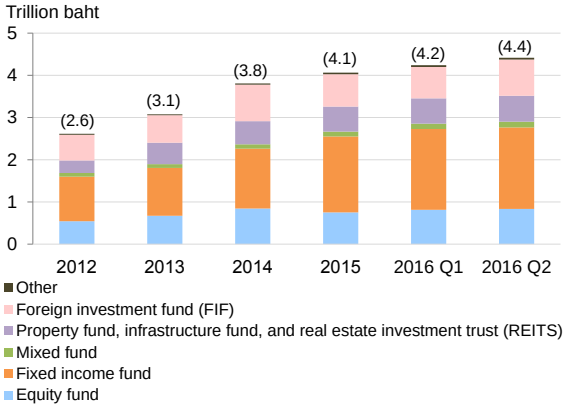
Investment in mutual funds continued to rise. Net asset value (NAV) under management of mutual funds grew in almost all fund categories in the second quarter of 2016, with the highest growth observed in fixed income funds including **deposit- like funds** such as **money market funds (MMFs)** and **daily fixed income funds (Daily FI)**

(Chart 3.21). Recently, investors shifted from MMFs, which focused on low risk and highly liquid short-term debt securities, to Daily FI in search for higher yields. As Daily FI could invest in long-maturity bonds and higher credit risk bonds, NAV of Daily FI had greater volatility linked to interest rate movements and changes in credit ratings of issuers. Investor confidence, if undermined, could lead to mass unit redemptions, and spark volatility in the money market. This issue needed to be monitored, together with promoting proper understanding among investors of the risks involved in each type of investment. At the same time, growth of **foreign investment funds (FIFs)** was also observed. FIFs recently turn to increase investment in countries which had higher sovereign credit ratings such as the United Arab Emirates, Luxemburg, Hong Kong, and Qatar, focusing mostly in investment-grade bonds.

As an alternative for those seeking yields higher than bank deposit rates, **savings cooperatives'** deposits and shares gained traction among savers, with growth sustained over the recent period **(Chart 3.22).** The main reason was that savings cooperatives' deposit rates and dividend yields were higher than those offered by commercial banks. The search-for-yield behavior was also exhibited by savings cooperatives, as evidenced by the increasing proportion of their equity assets with higher risks and expected returns. Nonetheless, equity-based assets accounted for only 1.5 percent of their total assets **(Chart 3.23).**

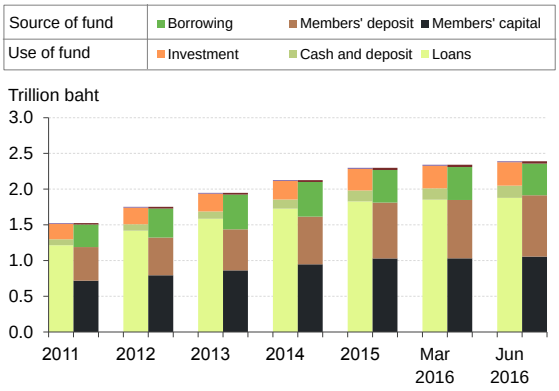
Overall, **systemic risks stemming from the search-for-yield behavior were still limited and signs of broad-based underpricing of risks were not present.** Nonetheless, the Committee would continue to monitor the search-for-yield behavior closely. This applied to both investors,

Chart 3.21 Net asset of mutual funds classified by asset class



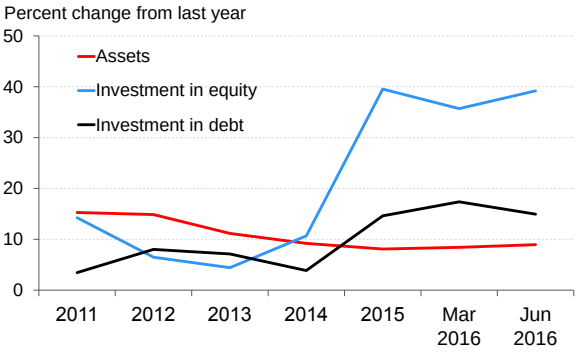
Note: *Data in () indicates total asset of mutual funds
Source: Association of Investment Management Companies, calculations by Bank of Thailand.

Chart 3.22 Saving cooperatives' use and source of fund



Source: Cooperative Auditing Department, calculations by Bank of Thailand

Chart 3.23 Growth in asset and investment of saving cooperatives



Source: Cooperative Auditing Department, calculations by Bank of Thailand

who continued to seek higher returns, and mutual funds that tended to invest in riskier assets such as equity, non-investment grade bonds and unrated bonds amid the low for longer environment. Vulnerabilities in the financial system that could accumulate as a result of financial institutions competing to extend loans to large corporates, together with concentration of loans in some business sectors, also warranted close monitoring.

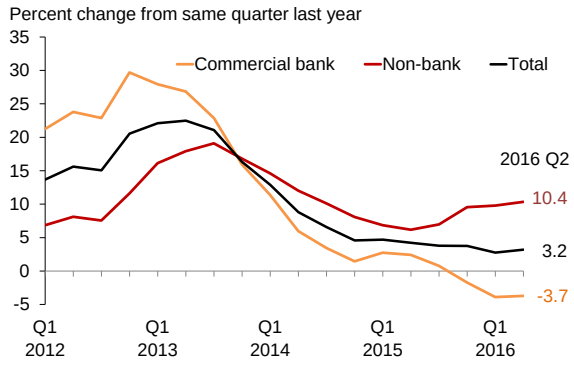
3.3 Non-financial sectors

Household sector

Households' debt serviceability continued to fall, as reflected by rising NPL levels. High debt levels and gradual economic recovery affected household economic and financial activity. However, slower growth in household debt and improved farm income would partly help improve the financial position of households going forward.

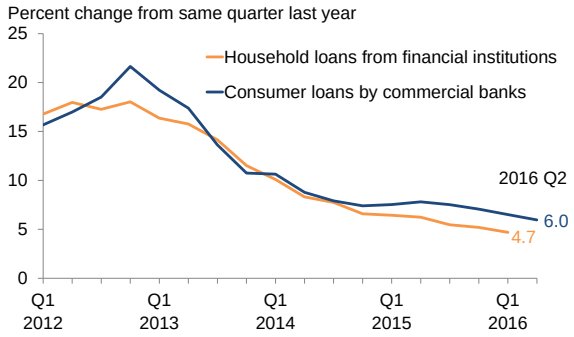
Households' debt serviceability decreased slightly, as reflected in the increase in the NPL ratio of consumer loans from 2.58 percent in the first quarter of 2016 to 2.60 in the second quarter, mainly due to lower credit quality of home loans. With weaker households' debt serviceability in the recent period prompting banks to be stricter in credit extension, some households shifted to borrowing from non-banks, particularly in the case of personal loans under regulation (PLR). Consequently, in contrast with the slowdown in commercial banks' credit extension, PLR growth accelerated **(Chart 3.24)**. Given that PLR issued by non-banks were mainly extended to low-income households, vulnerability in the household sector could accumulate in the future, which would need to be monitored closely.

Chart 3.24 Growth in personal loan under regulation



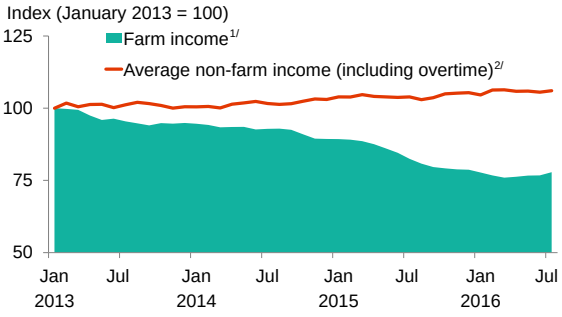
Source: Bank of Thailand

Chart 3.25 Growth in household loans issued by financial institutions nd consumer loans by commercial banks



Source: Bank of Thailand

Chart 3.26 Household income



Note: 1/ Seasonally adjusted, 12-month moving average
2/ Seasonally adjusted, 3-month moving average
Source: Office of Agricultural Economics, National Statistical Office; calculations by Bank of Thailand

Despite the outlook for slower growth of household debt (Chart 3.25) and gradual improvements in farm income due to expected increases in agricultural output and prices in line with global prices **(Chart 3.26),** high levels of household debt and uncertainties surrounding the economic recovery still constrained household economic and financial activity.

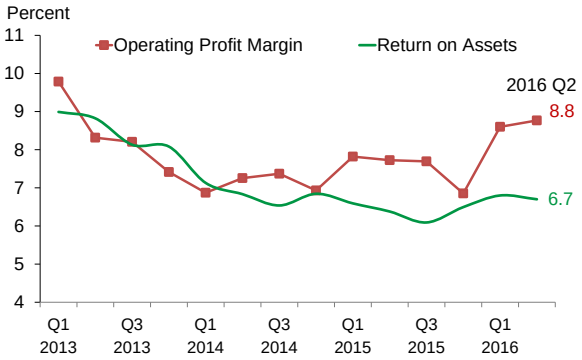
Business sector

In the second quarter of 2016, performance of non-financial companies listed on the SET slightly improved from the previous quarter. However, debt serviceability of SMEs that faced liquidity constraints would warrant close monitoring going forward.

Overall performance of businesses improved slightly from the previous quarter. Businesses' ability to generate revenues and profits improved, as reflected in the operating profit margins and returns on assets of most businesses **(Chart 3.27).** Increased profits in some sectors were driven by sales growth and effective cost management, notably in the real estate and wholesale and retail trade sectors. While the petroleum and petrochemical sector continued to experience sales contraction, profits improved thanks to inventory profits that were realized from higher oil prices **(Chart 3.28).**

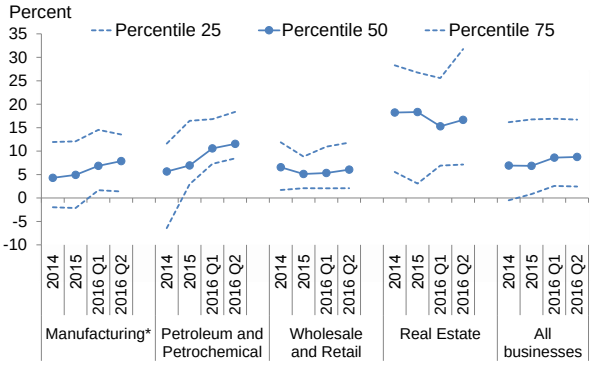
Business debt stabilized at low levels as reflected in the low median D/E ratio at 0.7. **Overall debt serviceability remained sound** with the median ICR at 5.8, well above the international standard of 1.5 **(Chart 3.29).** Businesses in most sectors were able to manage operating costs and reduce interest costs more effectively. This could be done by issuing debt instruments with lower funding costs than borrowing from banks. However, debt serviceability of some business sectors, such

Chart 3.27 Operating profit margin and return on assets



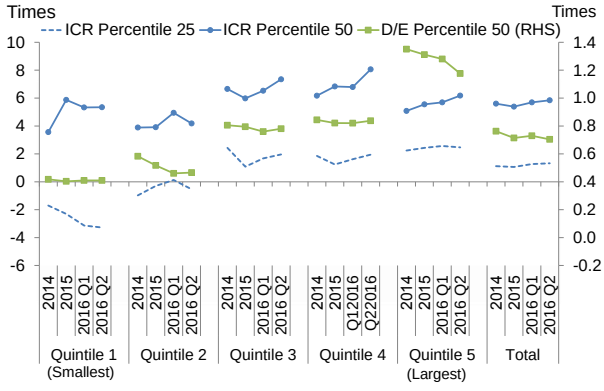
Note: Median estimates
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Chart 3.28 Operating profit market by sector



Note: *Exclude Petroleum and Petrochemical
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

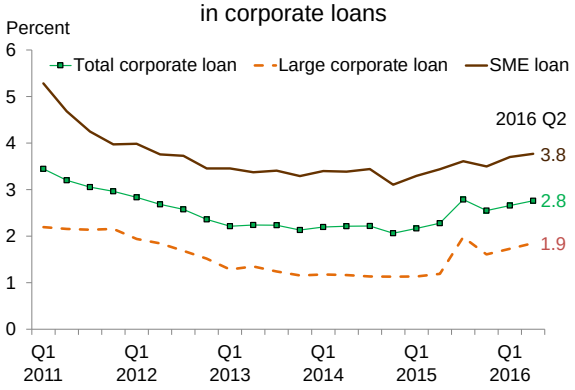
Chart 3.29 Interest Coverage Ratio (ICR)* and Debt to Equity (D/E) ratios classified by firm size



Note: *The 25th percentile reflects firms with low debt serviceability; ICR calculation is annualized; D/E is median estimate; Firms divided based on their asset size into 5 quintiles from smallest to largest.
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

as the mining industry, was impaired by the impact of continuing low prices of crude oil and coal. Risks in some sectors continued to warrant monitoring. Examples included those sectors that were sensitive to sluggish economic recovery, especially small enterprises with limited access to financing. Recent data on listed companies with assets in the two lowest quintiles indicated that they could be facing liquidity constraints due to persistent losses. Using these smaller companies as proxy for non-listed SMEs, it could be implied that debt serviceability of SMEs appeared to decline. This was consistent with the NPL ratio of SMEs at the end of the second quarter of 2016 which was on an upward trend **(Chart 3.30).**

Chart 3.30 Share of non-performing loans in corporate loans



Source: Bank of Thailand

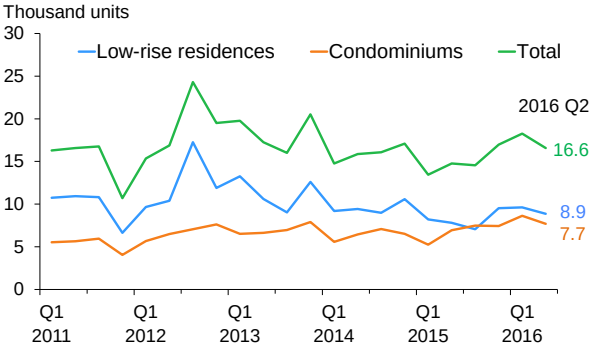
Real estate sector

The property market in Bangkok and its vicinities slowed down in the second quarter of 2016 after the government's stimulus measures expired. Risks of unsold stock of condominium units, especially projects along the mass transit lines, would warrant continuing monitoring.

The property market in Bangkok and its vicinities slowed down in the second quarter of 2016 after the government's real estate stimulus measures ended in April 2016, as reflected in the reduction of new housing loan approvals by commercial banks **(Chart 3.31).** On the supply side, developers also adjusted by launching fewer new projects and focusing instead on selling outstanding units so as to benefit from the government's measures, as reflected in the slowdown of new residential projects launched in Bangkok and its vicinities **(Chart 3.32).**

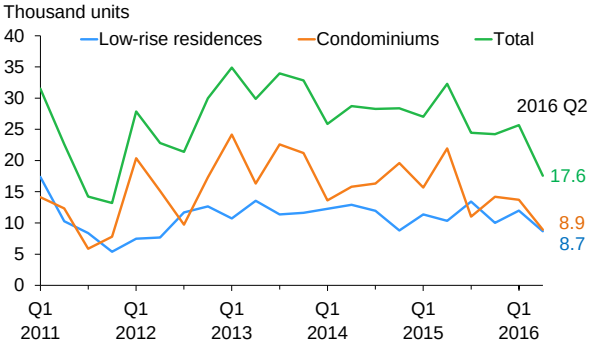
Demand was expected to slow down in the second half of 2016 compared with the first half as part of future property purchases had been brought

Chart 3.31 Residential units in Bangkok and its vicinity with approved mortgages by commercial banks



Source: Bank of Thailand

Chart 3.32 New residential projects launched in Bangkok and its vicinity



Source: Agency for Real Estate Affairs (AREA), calculations by Bank of Thailand

forward to benefit from the government's stimulus measures. On the supply side, the number of new projects launched gradually picked up in the second half of the year compared with the first half of the year, notably new condominium projects along the mass transit lines. Preliminary assessment indicated that the new supply was not a concern, given the few number of new projects launched in the first half. Moreover, the total number of new projects launched in 2016 appeared to be below levels in the past. However, risks associated with the increase in outstanding supply would warrant monitoring, especially if the new projects were located in areas with plenty outstanding condominium units, for instance units priced at 2-5 million baht along the Purple Line route and units priced over 5 million baht along the Green Line extension route **(Table 3.3).**

Table 3.3 Share of unsold condominium units in the first half of 2016

Price range (million baht)	City center*	Green Line route**	Purple Line route
Below 1	0%	9%	25%
1 - 2	16%	38%	28%
2 - 3	12%	18%	38%
3 - 5	24%	26%	29%
5 - 10	25%	40%	24%
Over 10	39%	44%	5%

Note: * Covering areas in Phaholyothin, Pathumwan, Sukhumvit, Rama 4, Yannawa, Silom and Phranakorn
** Bearing-Samutprakarn route
Source: AREA and calculations by Bank of Thailand

Additional issues to be monitored include implication of the Land and Buildings Tax Act and the mass transit line extension projects which will affect land and house prices. Land supply was anticipated to increase as the act would impose greater tax burdens on landowners. Extension of the mass transit lines to suburban areas would create new areas for development for property developers. As a result, land prices would not likely rise as sharply as in the past.

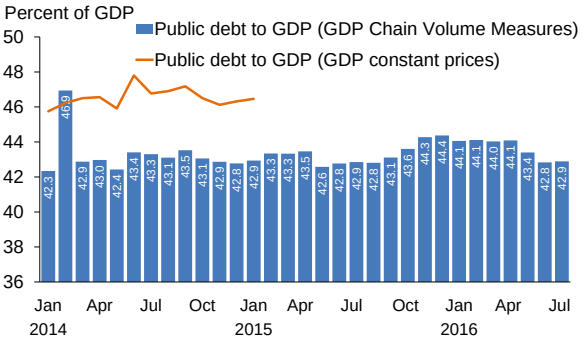
Fiscal sector

Fiscal stability remained sound, as reflected in public debt that remained below the fiscal sustainability threshold and the treasury cash balance that was sufficient to meet unexpected fiscal needs. Nonetheless, pressures on fiscal stability would likely increase from (1) additional borrowings needed for investment in transportation infrastructure projects and (2) future fiscal burden arising from recently approved stimulus measures.

Fiscal stability remained sound, as reflected in the public debt to GDP ratio that remained below the fiscal sustainability threshold of 60 percent. As of July 2016, the ratio stood at 42.9, slightly down from 44.1 percent in April 2016 **(Chart 3.33).** The decline was due to repayments of principal by the Financial Institutions Development Fund Phase II, B.E. 2545 (FIDF 3), Thai Airways International Public Company Limited, the Bank for Agricultural and Agricultural Cooperatives, together with bond redemptions by the Provincial Electricity Authority. In addition, the stronger baht lowered the remaining outstanding balance in baht terms. However, the central government continued to increase borrowing to finance its budget deficit, and to lend to state-owned enterprises for investment in (1) the Mass Rapid Transit Authority of Thailand's projects on the Green, Blue, and Purple Lines and (2) the State Railway of Thailand's projects to improve railways which were deemed unsafe for road transportation. Meanwhile, the treasury cash balance increased from the previous quarter. At the end of July 2016, the treasury cash balance stood at 248 billion baht, up from 206 billion baht at the end of April 2016. The increase was primarily attributed to transfer of receipts from the auction of 900 MHz spectrum licenses and digital TV licenses.

Going forward, pressures on fiscal stability were likely to increase due to (1) non-budgetary borrowing for investment in the transportation infrastructure projects under the Urgent Transportation Action Plan and (2) the additional fiscal burden from quasi-fiscal activities that were part of the economic stimulus packages, such as compensation for the interest rate differentials to specialized financial institutions taking part in the soft loan programs.

Chart 3.33 Ratio of public debt to GDP



Note: (1) Chart shows calendar years, (2) Official figures for debt-to-GDP ratio based on GDP chain volume measures are available only for February and March 2015. Figures prior to thi period are calculated by the Bank of Thailand

Source: Public Debt Management Office

Table 3.4 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2015	2015			2016			
			H1	Q3	Q4	Q3	Q4	Jul	Aug
1. Financial market sector									
Bond market									
Bond spread									
(10 years - 2 years)	1.3	1.1	0.9	1.2	1.1	0.7	0.5	0.5	N.A.
Equity									
SET Index (End of period)	1,497.7	1,288.0	1,504.6	1,349.0	1,288.0	1,407.7	1,445.0	1,524.1	1,548.4
Actual volatility (SET Index)[1/]	12.5	13.9	12.0	12.3	13.7	15.9	11.9	7.4	9.6
Price to Earnings Ratio (เท่า)	17.8	22.6	20.1	17.8	22.6	20.7	22.0	23.3	22.8
FX market									
Actual volatility (บาท) (%annualize)[2/]	4.0	5.1	4.5	5.2	6.1	4.4	4.6	3.8	4.1
Nominal Effective Exchange Rate (NEER)	104.3	108.5	110.8	106.6	105.9	106.7	105.1	105.6	105.7
Real Effective Exchange Rate (REER)	103.0	104.4	106.99	102.4	101.2	100.8	100.0	100.2	100.3
2. Financial institutions sector[3/]									
Minimum lending rate (MLR)[4/]	6.75	6.5	6.51	6.50	6.50	6.51	6.26	6.26	6.26
12-month fixed deposit rate[4/]	1.73	1.40	1.50	1.46	1.40	1.40	1.38	1.38	1.38
Capital adequacy									
Regulatory capital to risk-weighted asset (%)	16.8	17.4	16.8	17.3	17.4	17.5	17.5	N.A.	N.A.
Earning and profitability									
Net profit (billion baht)	214.1	192.3	103.7	50.0	38.6	51.2	50.9	N.A.	N.A.
Return on assets (ROA)	1.2	0.9	1.3	1.2	0.9	1.2	1.2	N.A.	N.A.
Liquidity									
Loan to deposit and B/E	95.7	97.0	96.5	97.3	97.0	94.8	96.4	N.A.	N.A.
3. Household sector									
Household debt to GDP (times)	79.9	81.6	80.6	80.8	81.6	81.1	N.A.	N.A.	N.A.
Financial assets to debt (times)	2.5	2.6	2.5	2.5	2.6	2.6	N.A.	N.A.	N.A.
Proportion of loans that are 3 months delinquent (%)									
Thai commercial banks :									
Consumer loans	2.4	2.6	2.6	2.8	2.6	2.6	2.6	N.A.	N.A.
Housing loans	2.2	2.4	2.4	2.6	2.4	2.6	2.7	N.A.	N.A.
Auto leasing	2.5	2.3	2.6	2.7	2.3	2.1	1.9	N.A.	N.A.
Credit cards	3.2	4.0	4.2	4.5	4.0	4.7	4.2	N.A.	N.A.
Other personal loans	2.5	2.7	2.8	2.9	2.7	2.7	2.7	N.A.	N.A.
4. Non-financial corporate sector[5/]									
Operating profit margin (%)	7.1	7.5	7.8	7.7	6.9	8.7	8.8	N.A.	N.A.
Debt to equity ratio (times)	0.8	0.7	0.7	0.7	0.7	0.7	0.7	N.A.	N.A.
Income coverage ratio (times)	6.0	5.1	5.2	4.8	5.4	5.5	5.8	N.A.	N.A.
Current ratio (times)	1.6	1.7	1.7	1.6	1.7	1.7	1.7	N.A.	N.A.

Table 3.4 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2015	2015			2016			
			H1	Q3	Q4	Q3	Q4	Jul	Aug
4. Non-financial corporate sector[5/] (Continues)									
Delinquency and NPL ratio									
Large business	1.1	1.6	2.0	2.0	1.6	1.7	1.9	1.9	N.A.
SMEs	3.1	3.5	3.4	3.6	3.5	3.7	3.8	4.0	N.A.
5. Real estate sector									
Demand									
The number of approved mortgages from banks									
(Bangkok and its vicintly)	62,839	59,667	26,503	15,770	17,394	15,487	15,611	2,981	N.A.
Single-detached and semi-detached houses	15,694	13,152	6,384	3,115	3,653	3,511	3,503	681	N.A.
Townhouses and commercial buildings	21,764	19,210	9,199	4,812	5,199	5,092	5,410	1,026	N.A.
Condominiums	25,381	27,305	10,920	7,843	8,542	6,884	6,698	1,274	N.A.
Supply									
The number of new openings (Bangkok and its vicintly)	111,211	107,988	59,316	24,436	24,236	25,667	17,584	12,138	N.A.
Single-detached and semi-detached houses	18,933	17,637	6,354	7,388	3,895	4,655	2,918	940	N.A.
Townhouses and commercial buildings	26,980	27,518	15,349	6,034	6,135	7,308	5,735	3,859	N.A.
Condominiums	65,298	62,833	37,613	11,014	14,206	13,704	8,931	7,339	N.A.
Housing price index[5/]									
Single-detached houses (including land)	127.9	129.3	127.7	129.6	129.3	129.4	129.4	133.7	N.A.
Townhouses (including land)	133.5	137.5	133.7	134.7	137.5	138.8	138.8	139.1	N.A.
Condominiums	151.1	160.9	152.1	156.4	160.9	160.0	160.0	162.1	N.A.
Land	145.6	168.8	151.9	157.8	168.8	169.5	169.5	173.7	N.A.
6. Fiscal sector									
Public debt to GDP (%)	46.3	44.4	42.8	43.1	44.4	44.0	42.8	42.9	N.A.

[1/] Annualized standard deviation of return

[2/] Daily volatility (using exponentially weighted moving average me

[3/] Based on data of all commercial banks

[4/] Average value of 4 largest Thai commercial banks

[5/] Only listed companies on SET (median); with revisions to data and methodology.

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